

Terrorists Win

Ace! Totally Dubbed killed the entire enemy team.

\$4950

Killer Hz

100

0

Cyber Mercenary
30+ years



**CISO / Co-
Founder**



DEFCON 23



I WILL KILL YOU

DEFCON 24

Q&A: Meet the hacker that can overthrow a government.

Planning a coup? Skip the guns and bombs and hire a team of hackers instead.



Alina Simone [Follow](#)
Aug 29, 2016 · 8 min read



**HOW TO OVERTHROW
A GOVERNMENT**

Where have all the good Terrorists gone ?



GLOBAL TERRORISM INDEX 2022

MEASURING THE IMPACT OF TERRORISM

2022 National Intelligence
report had Terrorism as

2016¹ China
National Intelligence
report had Terrorism down 39%
as
the world's biggest risk.
Deaths from Terrorism down 39%
report had Terrorism at #1 as
(MENA) in the last 3 years.

5. Health and Security
Terrorists Attacks in the West down
68% since 2018.
6. Climate Change

7. .

8. .

9. .

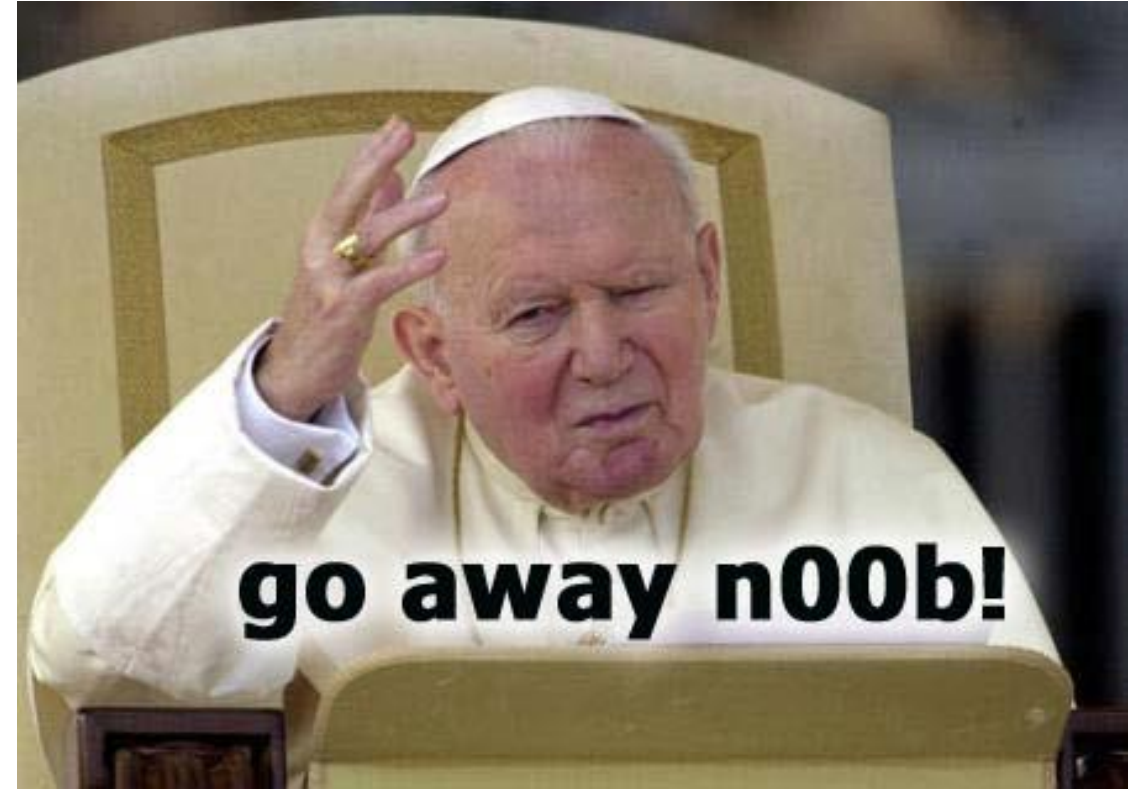
Footnote: Terrorism



Priest Installs Cellphone Jammer in Church to Decrease Interruptions During Mass

🕒 December 22, 2014 🧑 Item Media

📁 EMC, EMI, News



Vatican installs jammers to the Sistine Chapel to protect next Pope announcement.



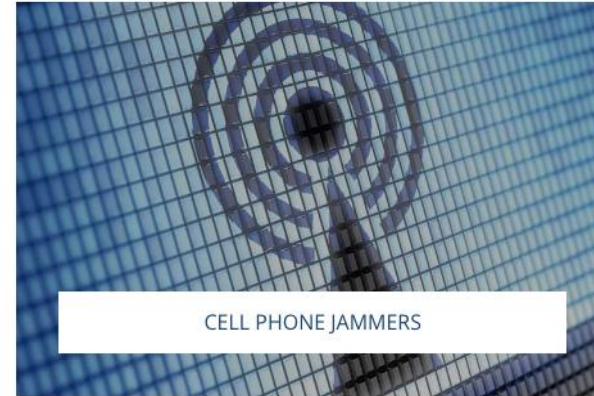
Thieves using 'jammers' to block alarms

Rise in burglars with high-tech devices which disrupt intruder alerts and phone signals during break-ins

MONITORED ALARM systems installed in homes, shops and businesses are being "blocked" by organised crime gangs using electronic devices.

The devices, known as GSM jammers, emit a powerful signal that blocks intruder alerts sent by alarm systems to monitoring stations.

Jammers



CELL PHONE JAMMERS



IED (BOMB) JAMMERS



SATELLITE JAMMERS

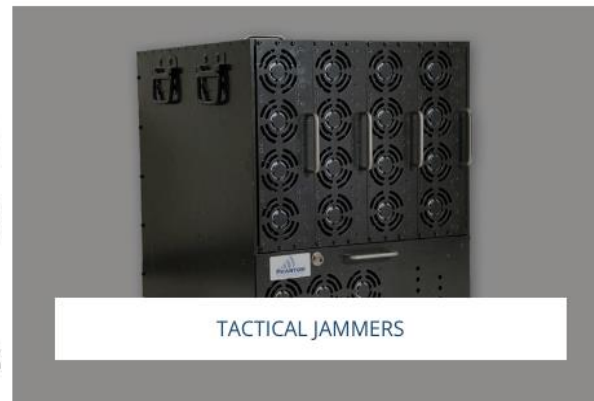


RESPONSIVE JAMMER



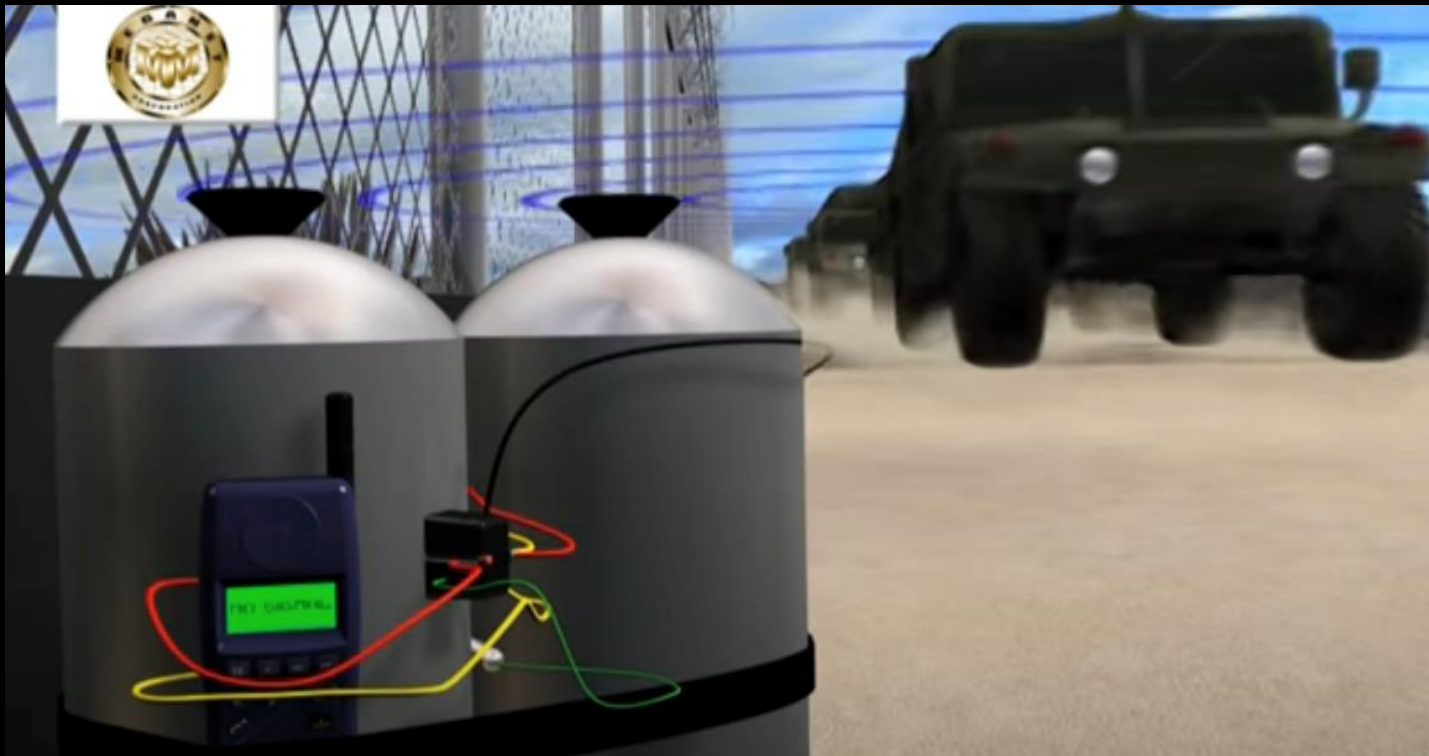
TACTICAL JAMMERS

3G
4G
LTE
WiFi
GPS
LO/JACK
UHF
433 MHz

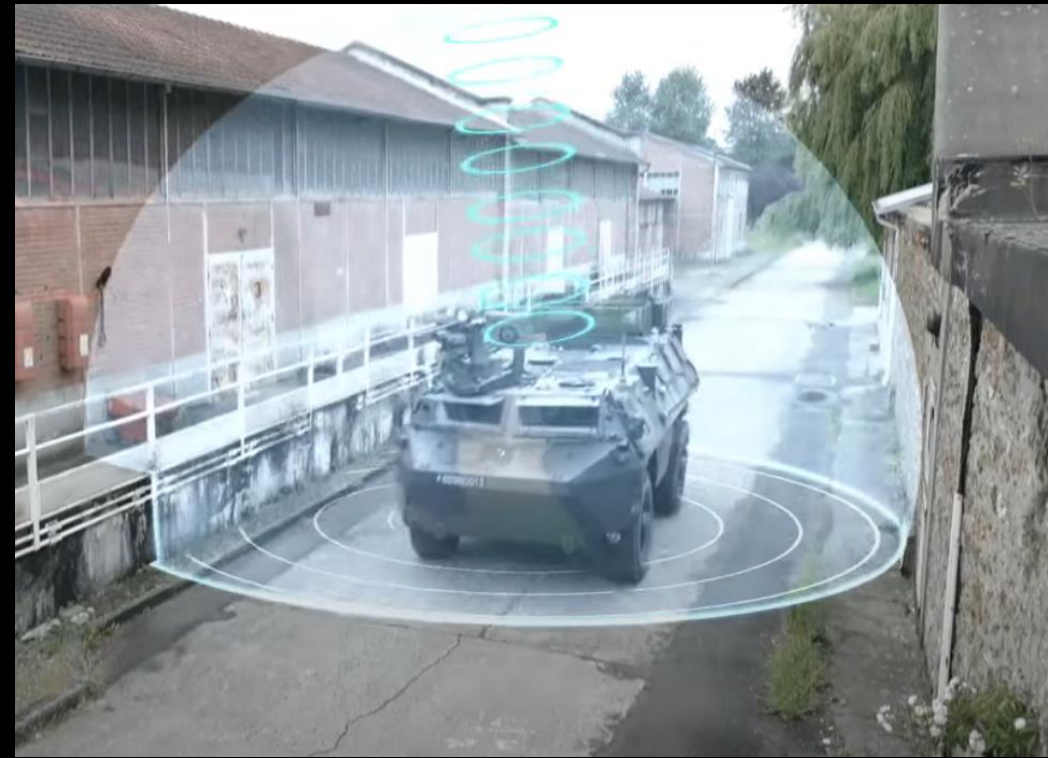


COUNTER SURVEILLANCE EQUIPMENT

Humvee Jammer



Thales Eclipse



Jammer Protection



Manufacturers

Raytheon

THALES

BAE SYSTEMS

NORTHROP GRUMMAN

Counter IED market is a 2-billion-dollar Industry

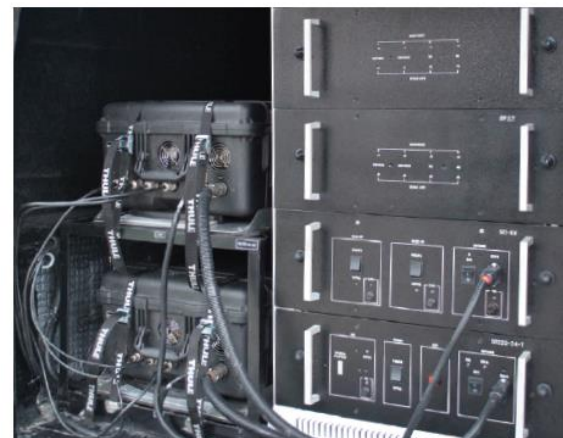
USMC Counter Radio-Controlled Improvised Electronic Warfare (CREW) 3.1 Thor III

System Characteristics:

TAMCN: A0326
NSN: 5865-01-580-4854
Technology: Digital
Spectrum: Classified
Orientation: Adjustable position for Antennas
Mobility: Man portable / Dismounted CREW
Power: 18 to 32VDC, 2.5A maximum (Low)
18 to 32VDC, 4.5A maximum (Mid)
18 to 32VDC, 5.0A maximum (High)
Distance: Classified
Operational Mode: Standby – System is idle, no scanning or jamming
Operate – System is scanning and jamming as programmed.
Encryption: N/A
Quantity Fielded: 0
AAO: 823 systems + spares (Pending SON)
336 systems (Original AAO)
+ 50 Spares per SONs

Military grade jammers

MCSV-3500W High power convoy jammer with communication windows setting



FEATURES:

- Over 1000W power at antenna output
- Independent 13 main channels covering 20-3500 MHz with 650W
- Additional 6 channels for cellular bands of total 358W
- 38 communication windows programming ability
- AC mains or vehicle power supply
- Suppression of entire bands without gaps
- Modular system allows customization on request

The **MVCS-3500W jamming system** is the military grade solution that can be used as stationary or convoy protection unit with ability of programming communication windows and channels on/off settings.

The MCSV-3500W provides minimum continuous wave signal of 1008W power on antenna output. The jamming range in between 20-3500 MHz is divided without any gap by 13 main channels and 6 additional channels especially for cellular communication frequency bands.

An operator can program up to 38 frequency windows within 13 main channels. The windows quantity and width depend on the channel.

The delivery set includes antennas for the whole frequency range. The power supplied via 22-30V on-board network. The power consumption is not more than 5kW. The MCSV-3500W can be stationary installed and connected to the 380V AC mains.

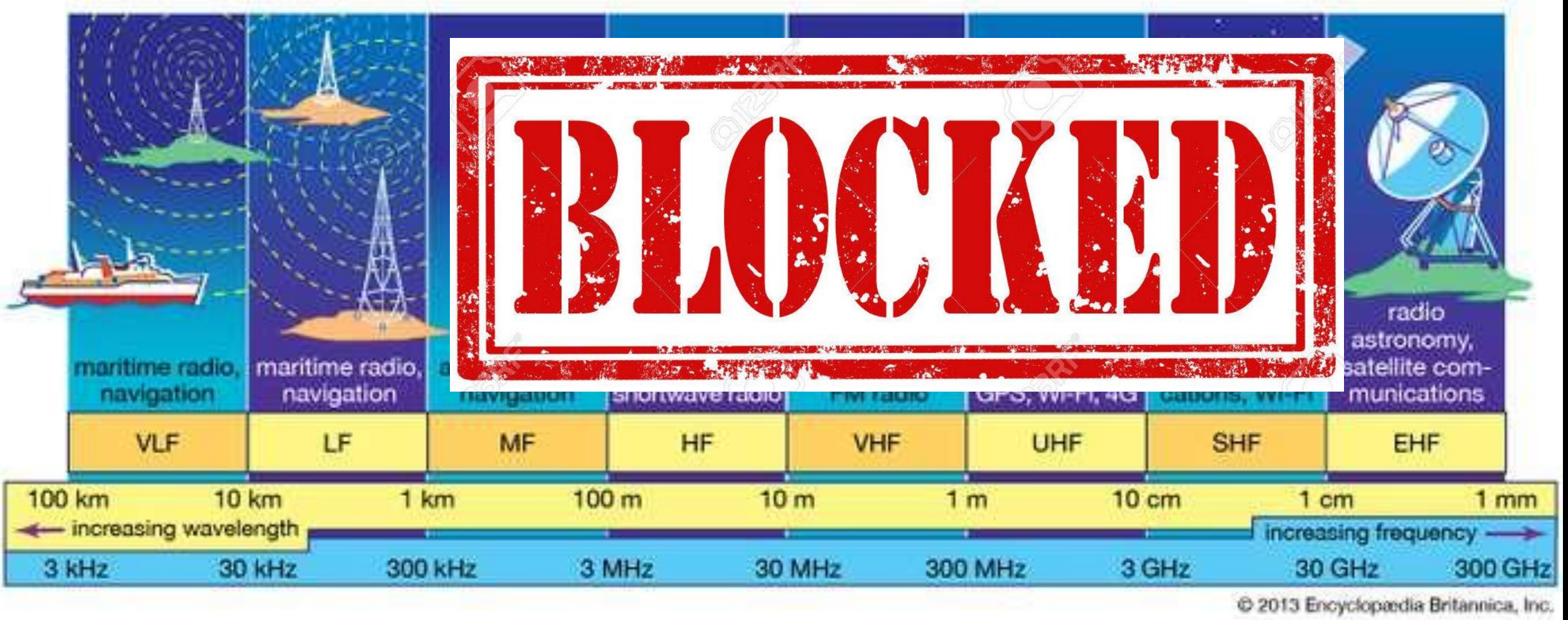
The device employs the maintenance control algorithm that activates the alarm in case of any operation failure or wire disconnection will automatically turn off the amplifier.

The delivery set includes all the antennas and installation kit. The manufacturer's team can install the jammer on the customer request.

PRODUCT SPECIFICATIONS

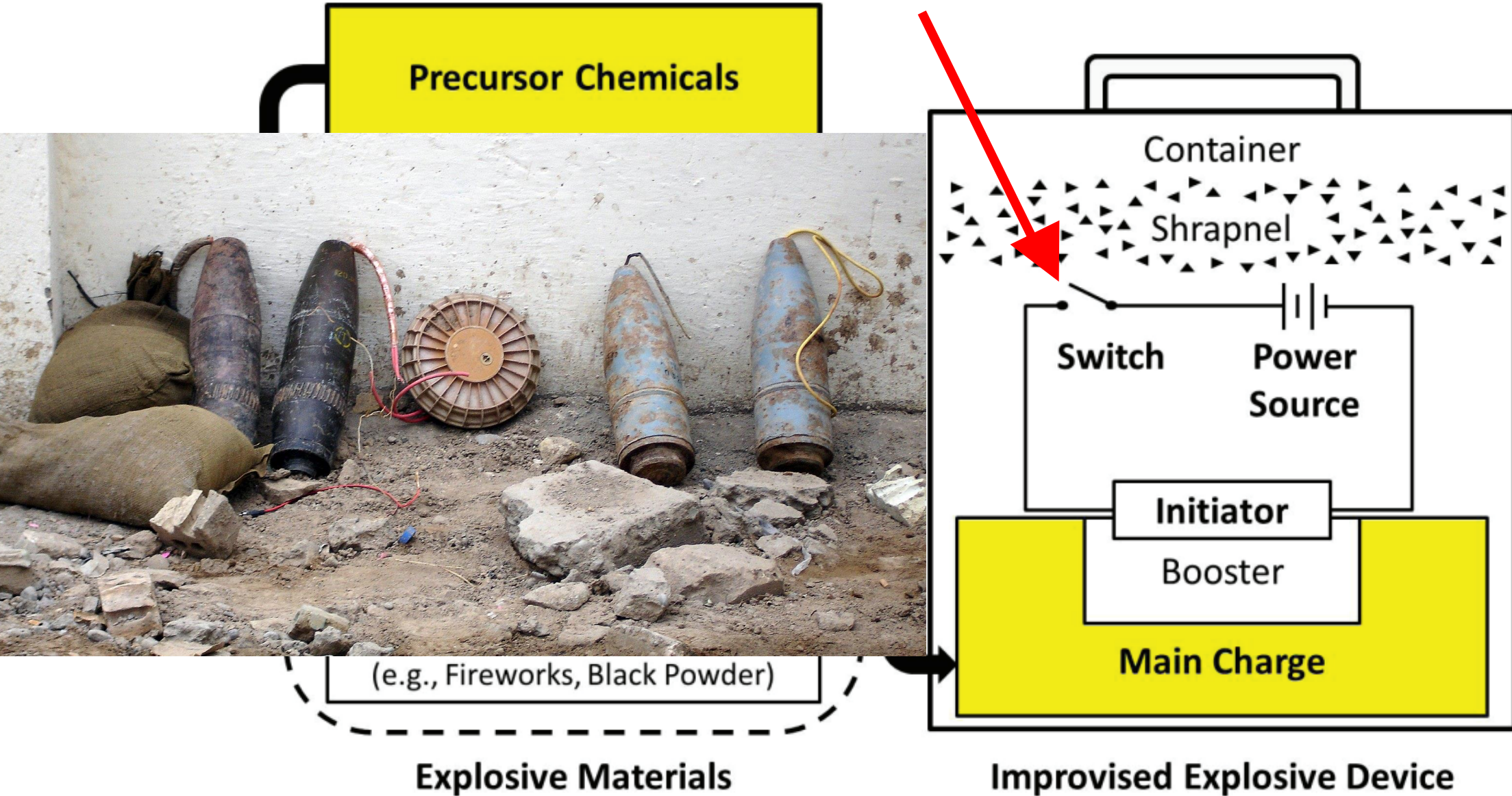
Operating frequency bands / Min output power	Total coverage of 20-3500 MHz by divided 19 channels					
	Ch1	20-60 MHz	60W	Ch11	1700-2000 MHz	50W
	Ch2	60-130 MHz	60W	Ch12	2000-2700 MHz	15W
	Ch3	130-180 MHz	100W	Ch13	2700-3500 MHz	15W
	Ch4	180-250 MHz	60W	Ch14	791-894 MHz	100W
	Ch5	250-400 MHz	100W	Ch15	925-960 MHz	100W
	Ch6	400-500 MHz	100W	Ch16	1805-1880 MHz	60W
	Ch7	500-750 MHz	40W	Ch17	2110-2170 MHz	60W
	Ch8	750-1000 MHz	50W	Ch18	2400-2500 MHz	25W
	Ch9	1000-1400 MHz	50W	Ch19	2500-2700 MHz	13W
	Ch10	1400-1700 MHz	50W			
Total output power	1008W					

Jammers

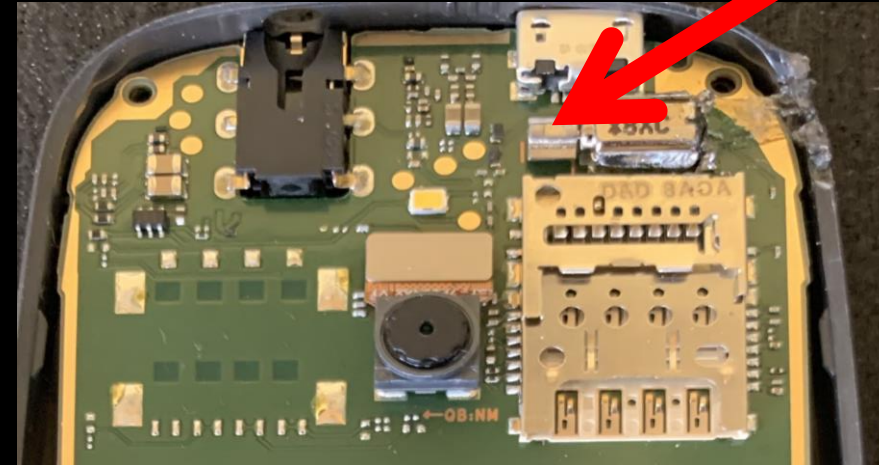
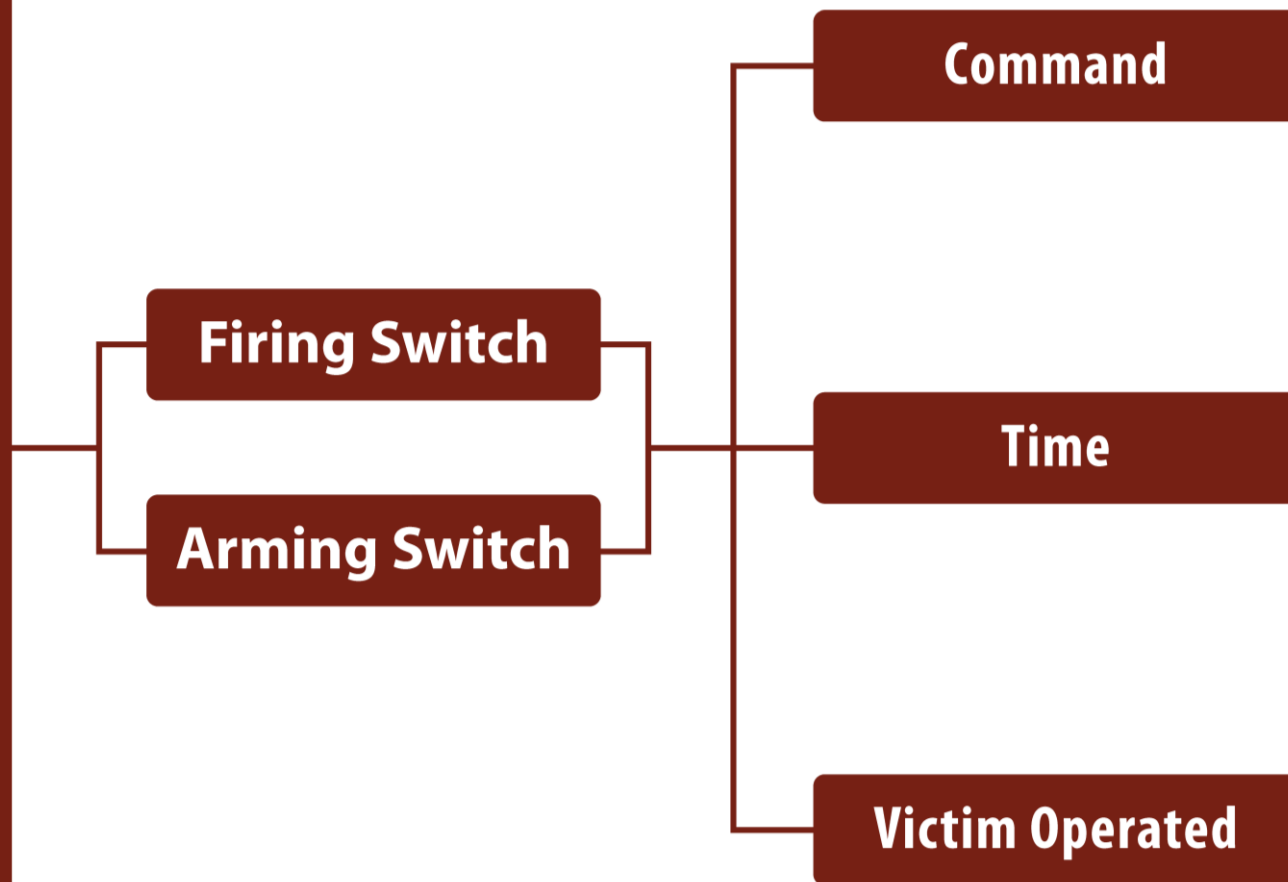


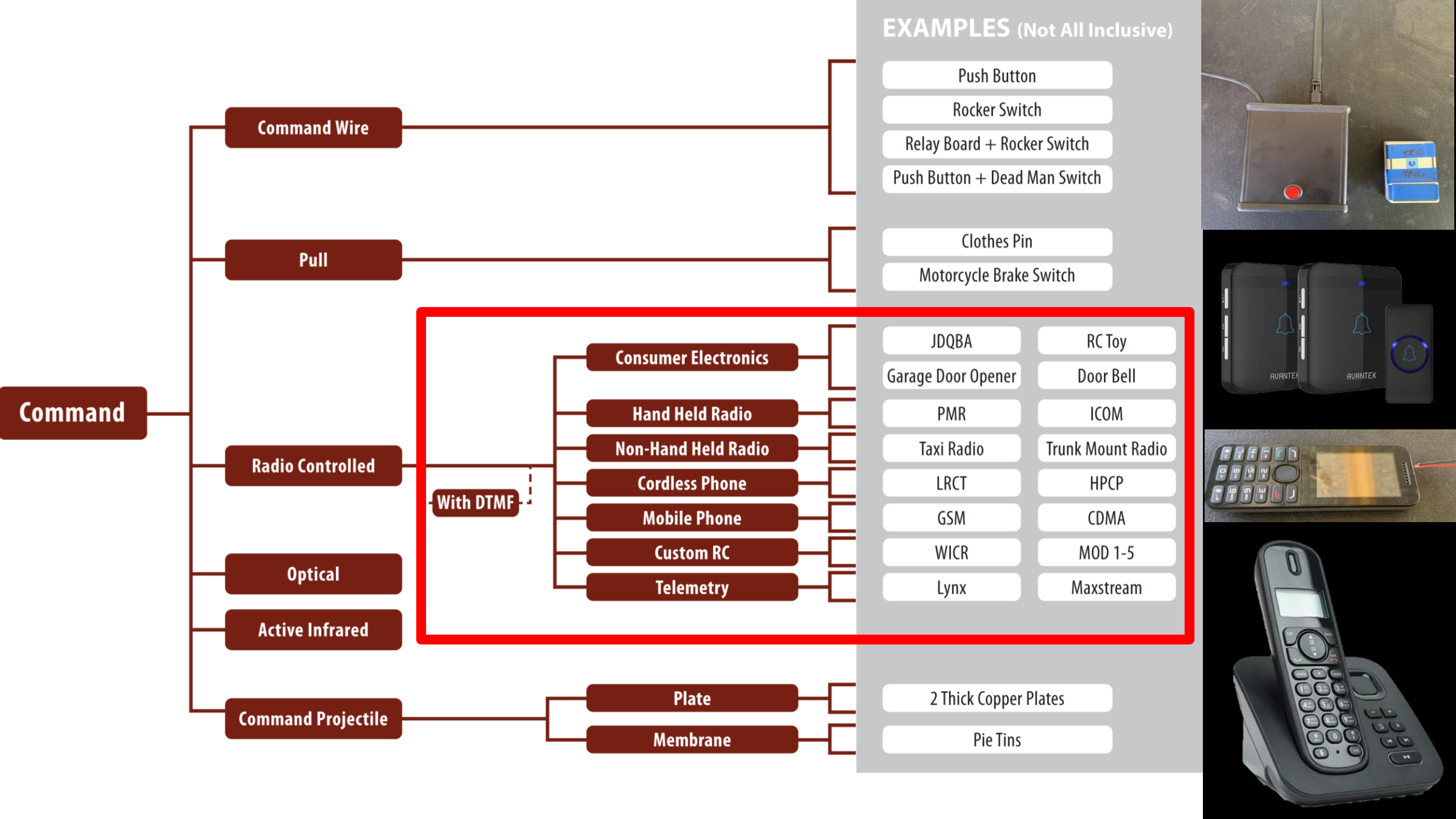
20MHz – 6500 MHz

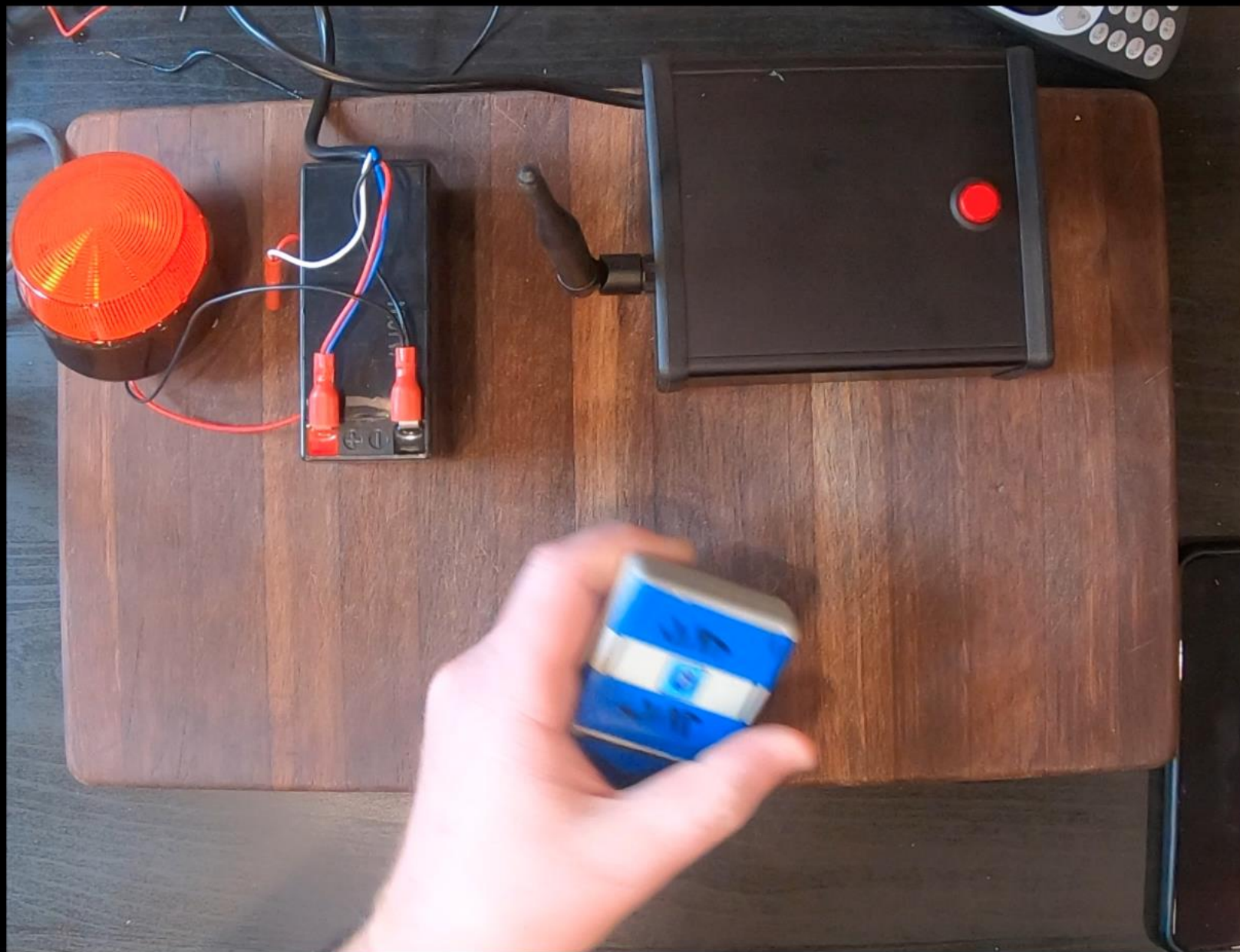
IED Overview



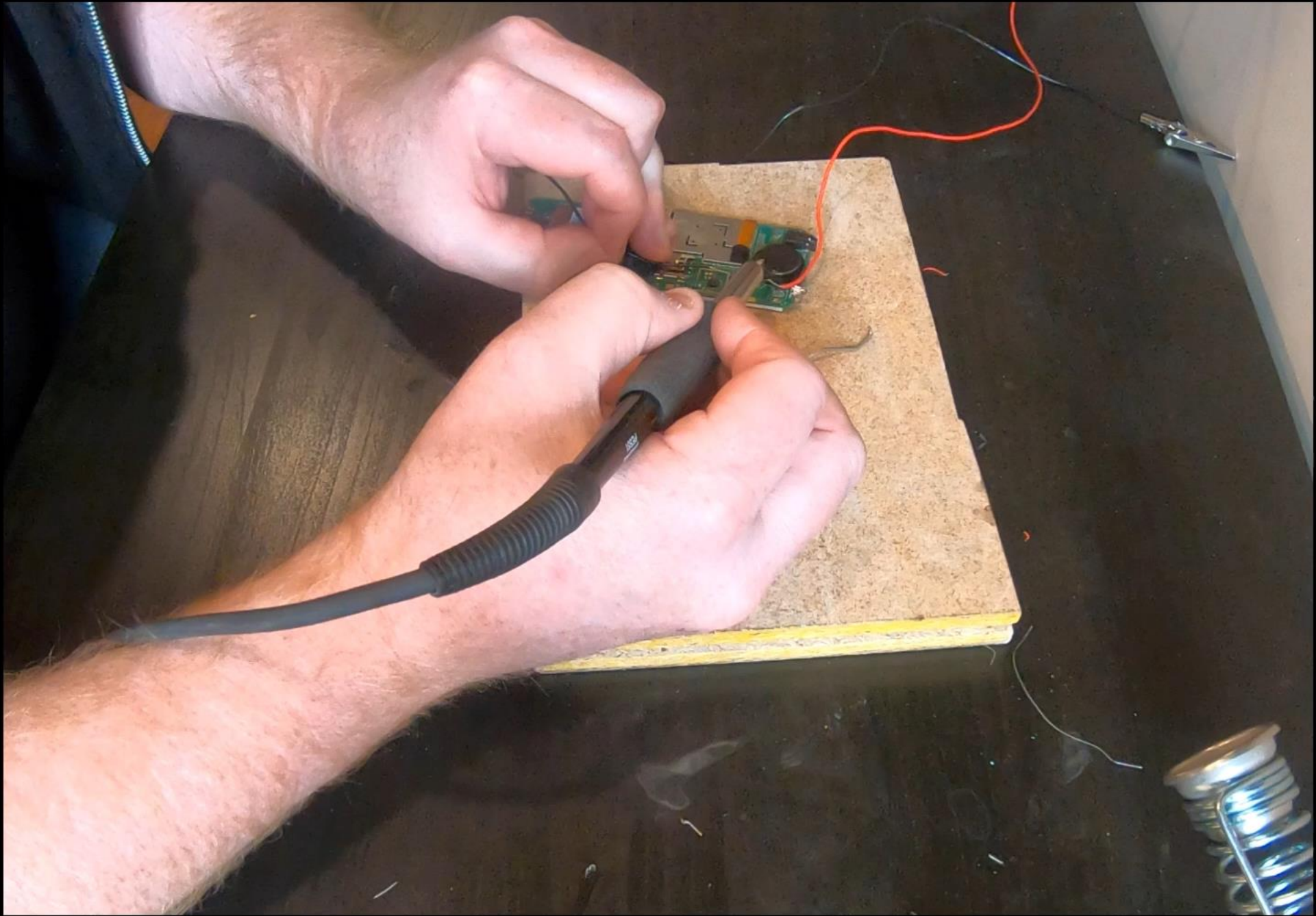
SWITCH







Garage door opener : Frequency 433 MHz

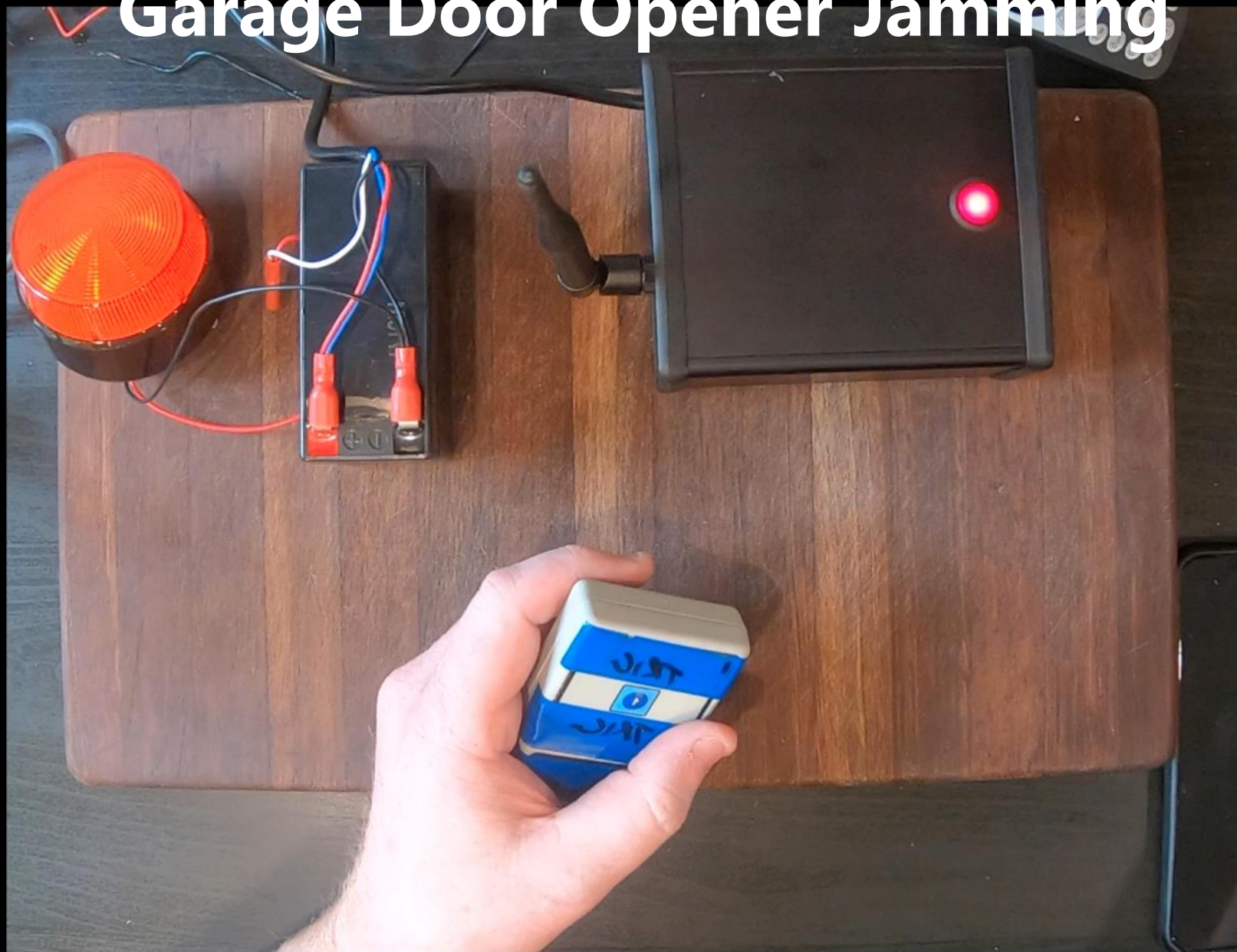


Cell Phone Switch Speaker-Vibrator Soldering



Remote Control IED (RCIED) Electronic Counter Measures (ECM)

Garage Door Opener Jamming



12 Channel Jammer Blocking GSM, Wifi, Bluetooth, 433 MHz and others.

Cell Jamming

Cell
Providers

WI-FI/GSM



Mobile Phone: Wifi, 4G, 3G

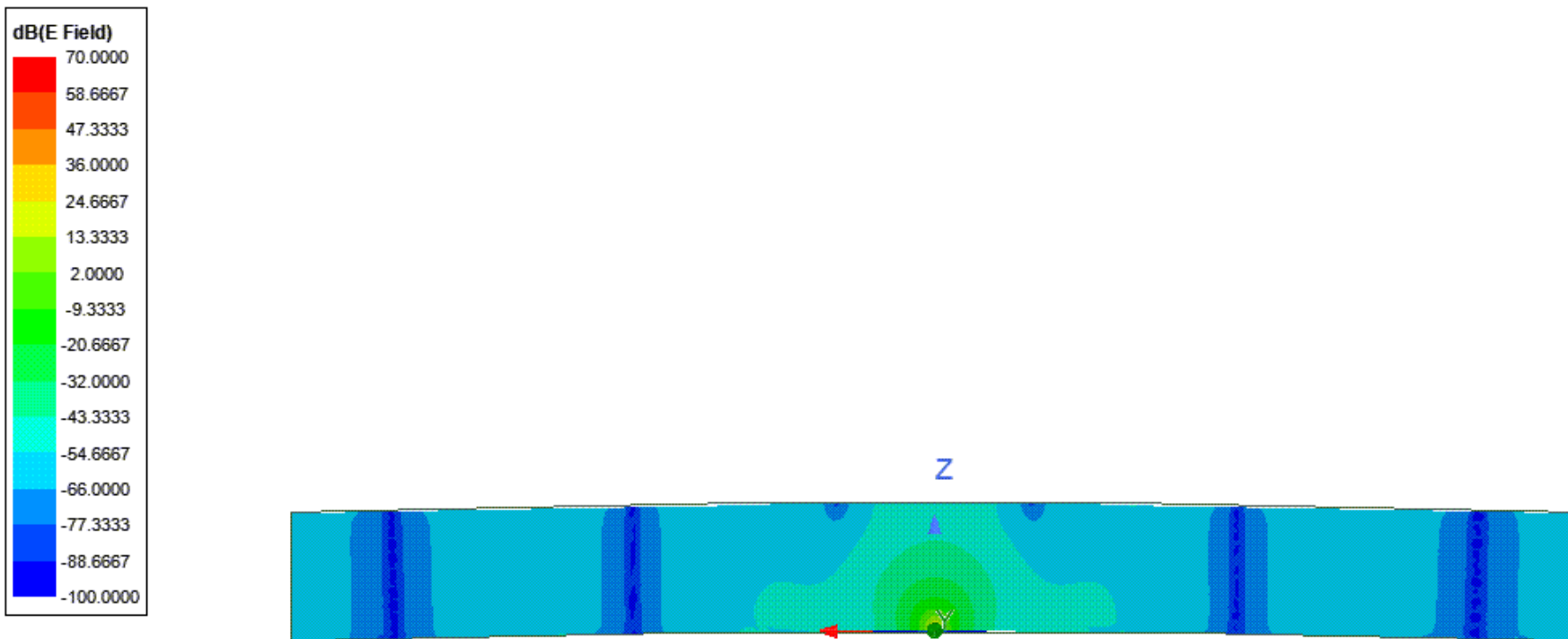


Jammer Gaps ?

Name	Symbol	Frequency Range	Wavelength
Extremely Low Frequency	ELF	3 Hz - 30 Hz	10,000 km – 100,000 km
Super Low Frequency	<u>SLF</u>	30 Hz - 300 Hz	1,000 km – 10,000 km
Ultra Low Frequency	ULF	300 Hz - 3 kHz	100 km – 1,000 km
Very Low Frequency	<u>VLF</u>	3 kHz - 30 kHz	10 km – 100 km
Low Frequency	<u>LF</u> or <u>LW</u>	30 kHz - 300 kHz	1 km – 10 km

BLOCKED

VLF / ULF / ELF Military



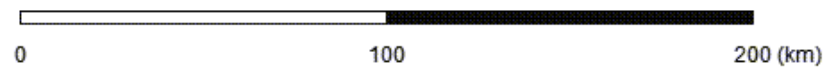
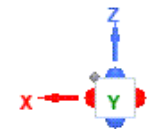
Ansys

Magnetosphere

Ionosphere

Atmosphere

Earth



Ultra Low Frequency (ULF)

Range: 300 Hz – 3 kHz

Wavelength = 1000 to 100km

2 kHz = Wavelength is 150Km

Antenna Requirements 2 kHz

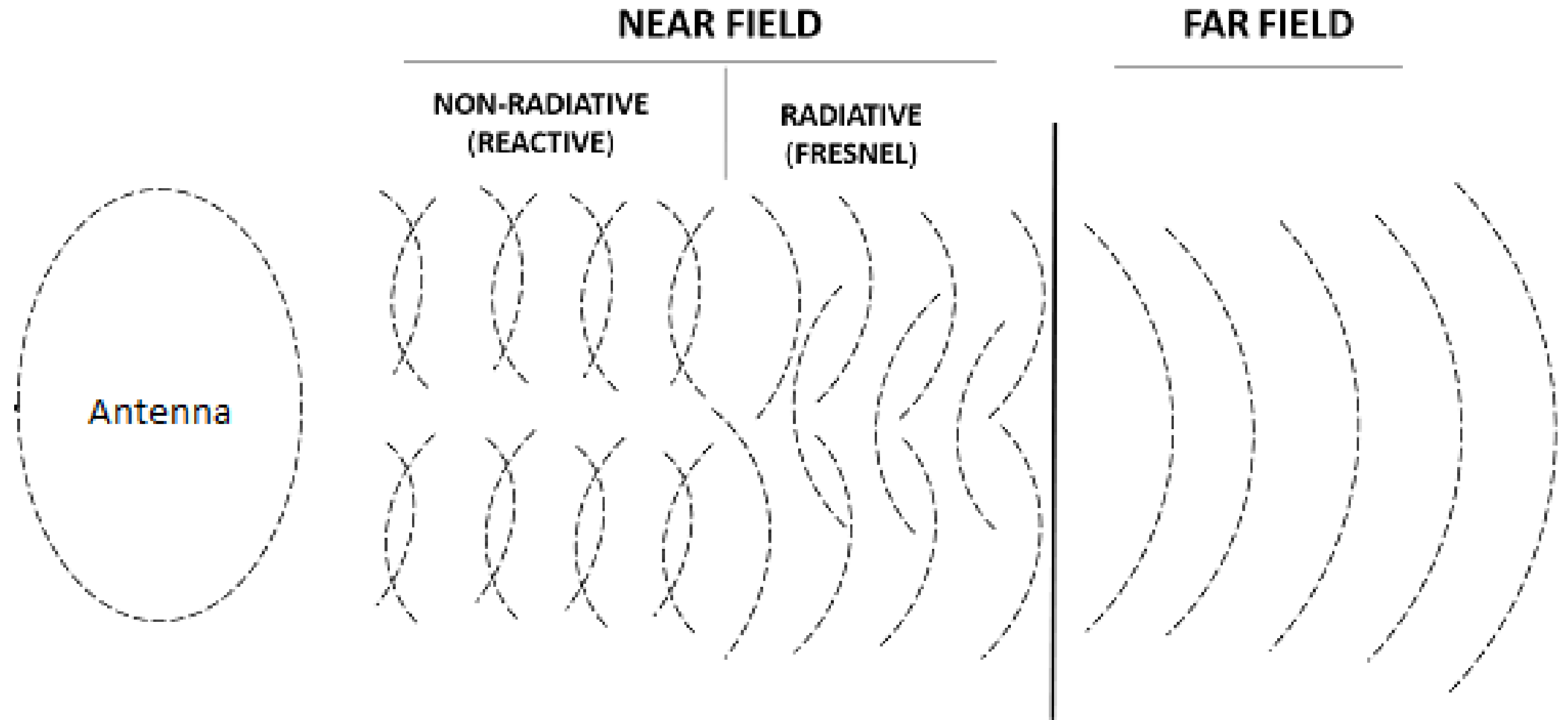
- **$\frac{1}{2}$ Wavelength = 75km**

Travels through the Earth and Sea Water

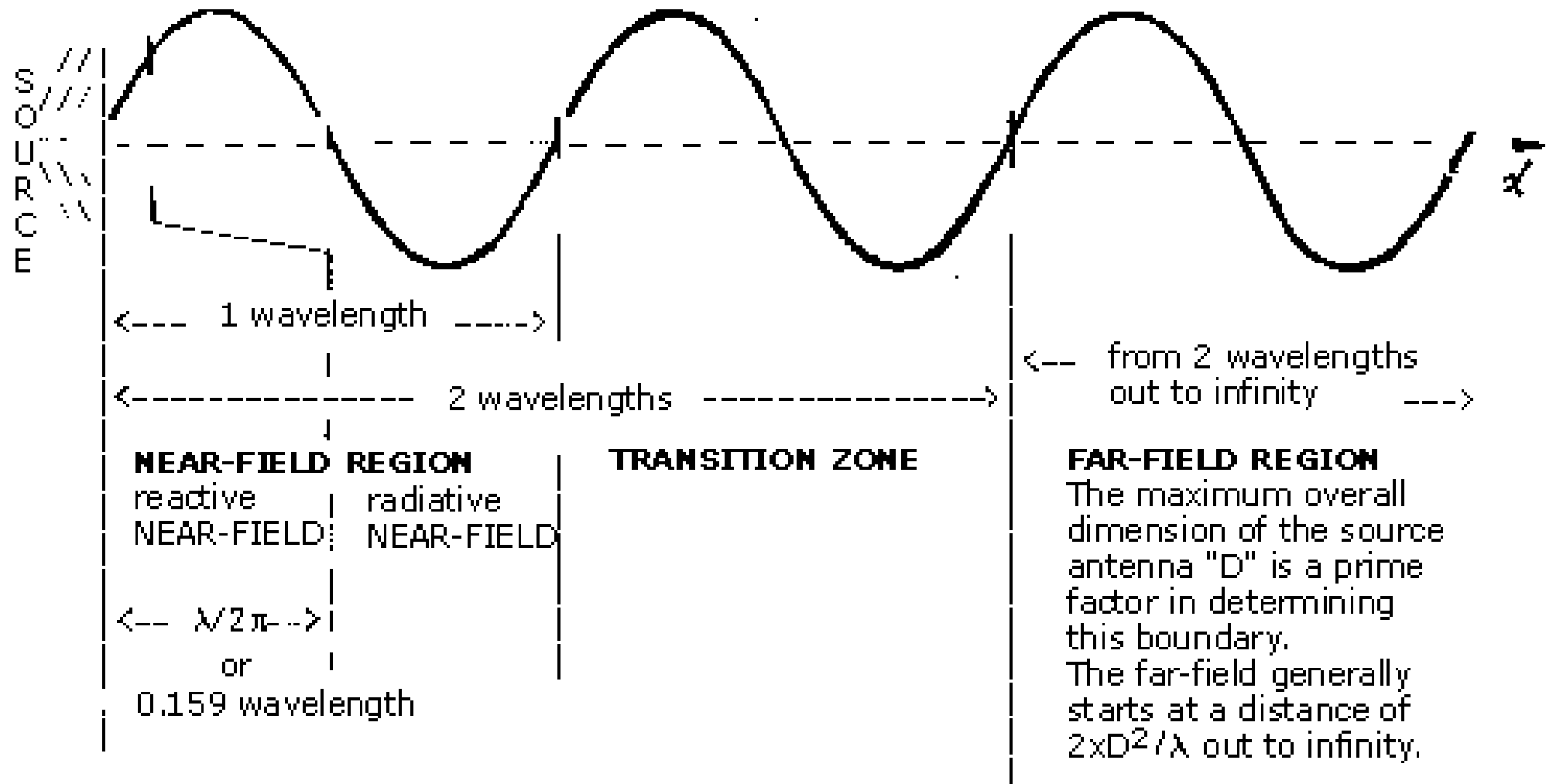
Near Field vs Far Field



Near Field vs Far Field

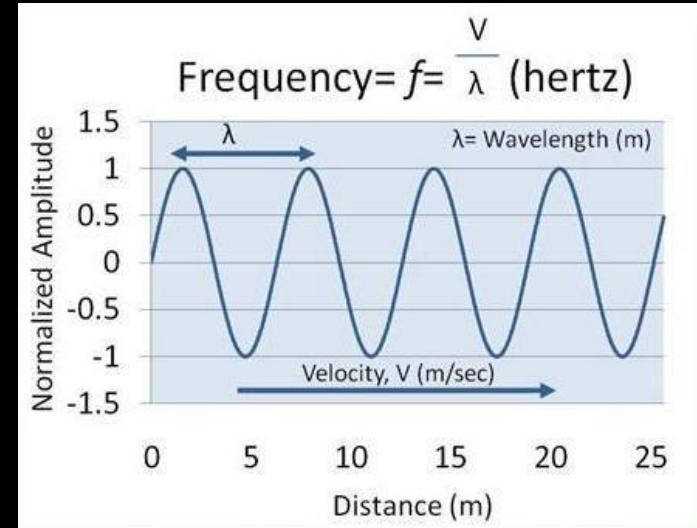
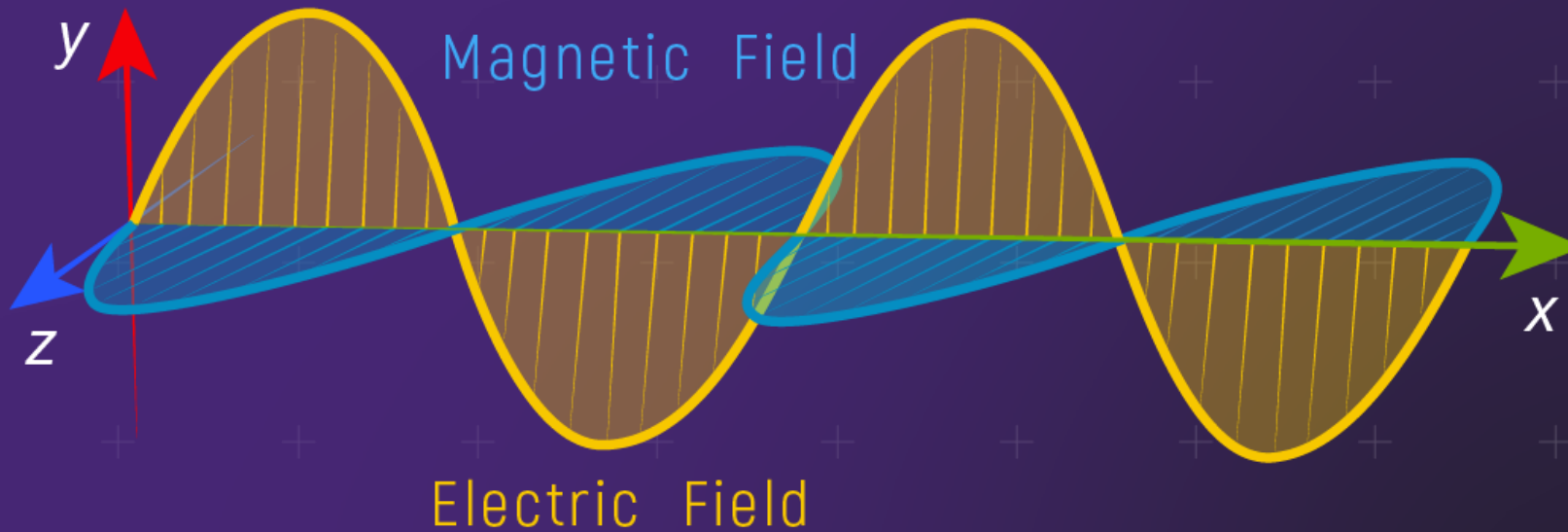


Near Field & Far Field Concept



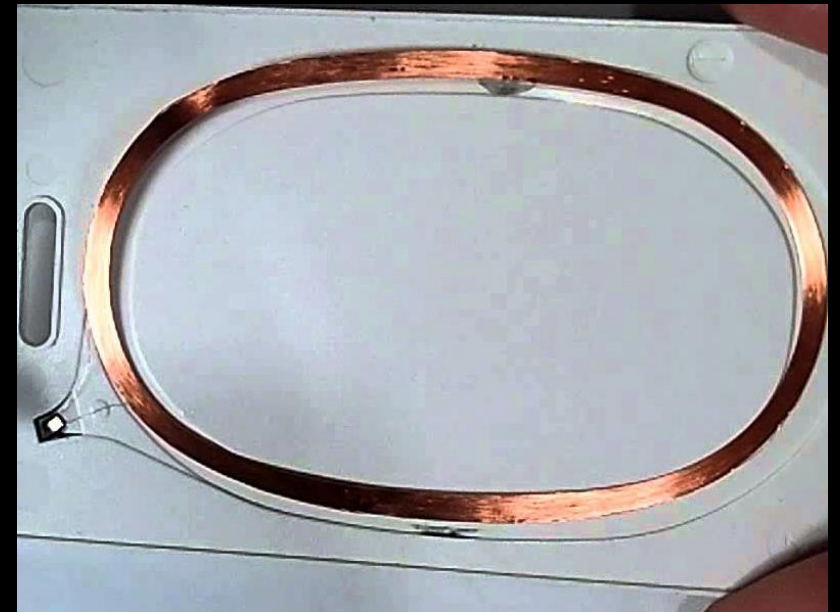
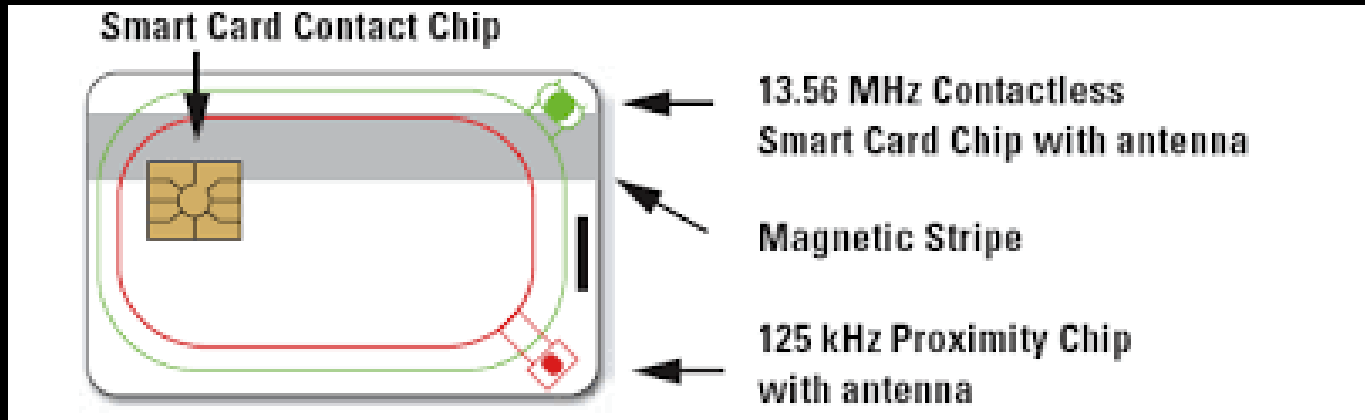
Radio Waves – Far Field

ELECTROMAGNETIC WAVE



Wavelength = Distance a wave take to complete a cycle.

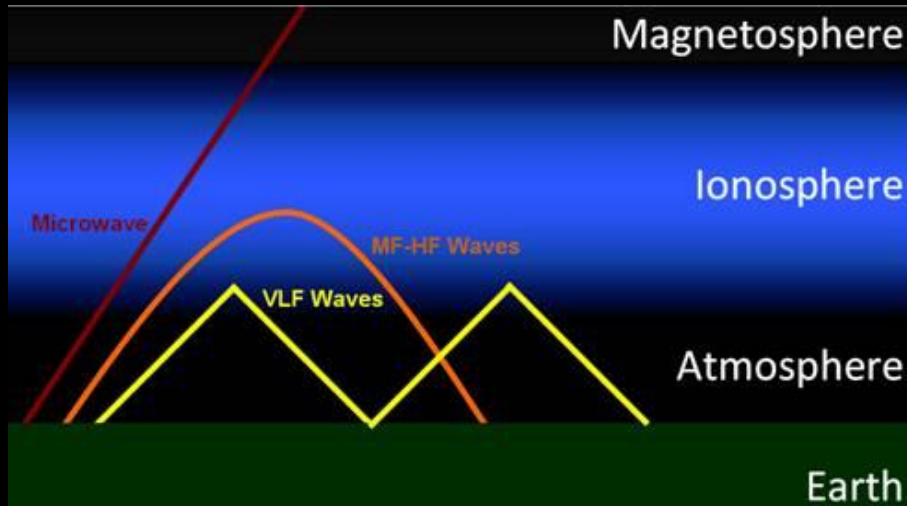
Frequency = How many repeating wave cycles per second.



NFC Range



13.56 MHz
Wavelength : 22.11 Meters
Near Field Range: 1-5 cm



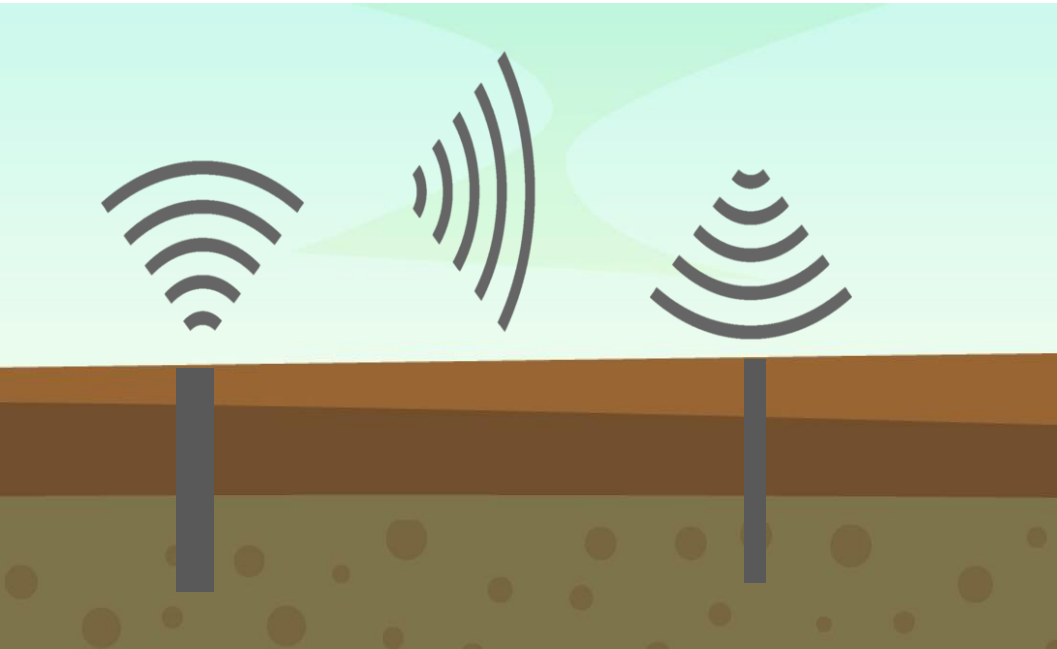
2 kHz
Wavelength : 150 kilometers
Near Field Range: 75 kilometers

**How do we create a
VLF/ULF
Wavelength**

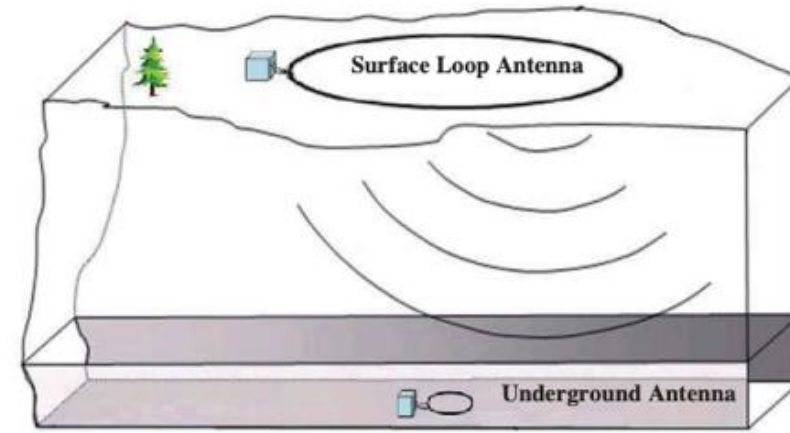
without the massive antenna?

VLF / ULF – Earth Mode

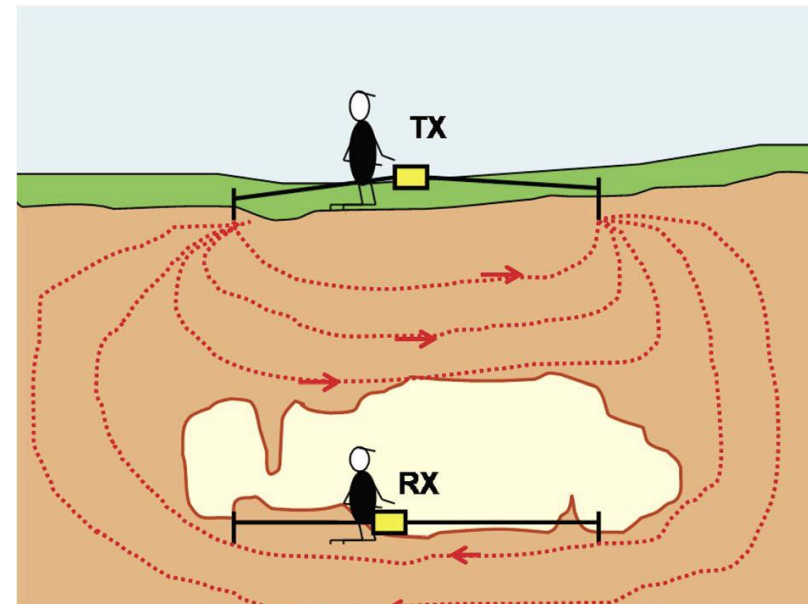
Earth Mode Methods



**Earth Current Communication
(up – over and down)**



**Deep Strata –
Mining Comms
PED**



**Conduction
Current Mode**

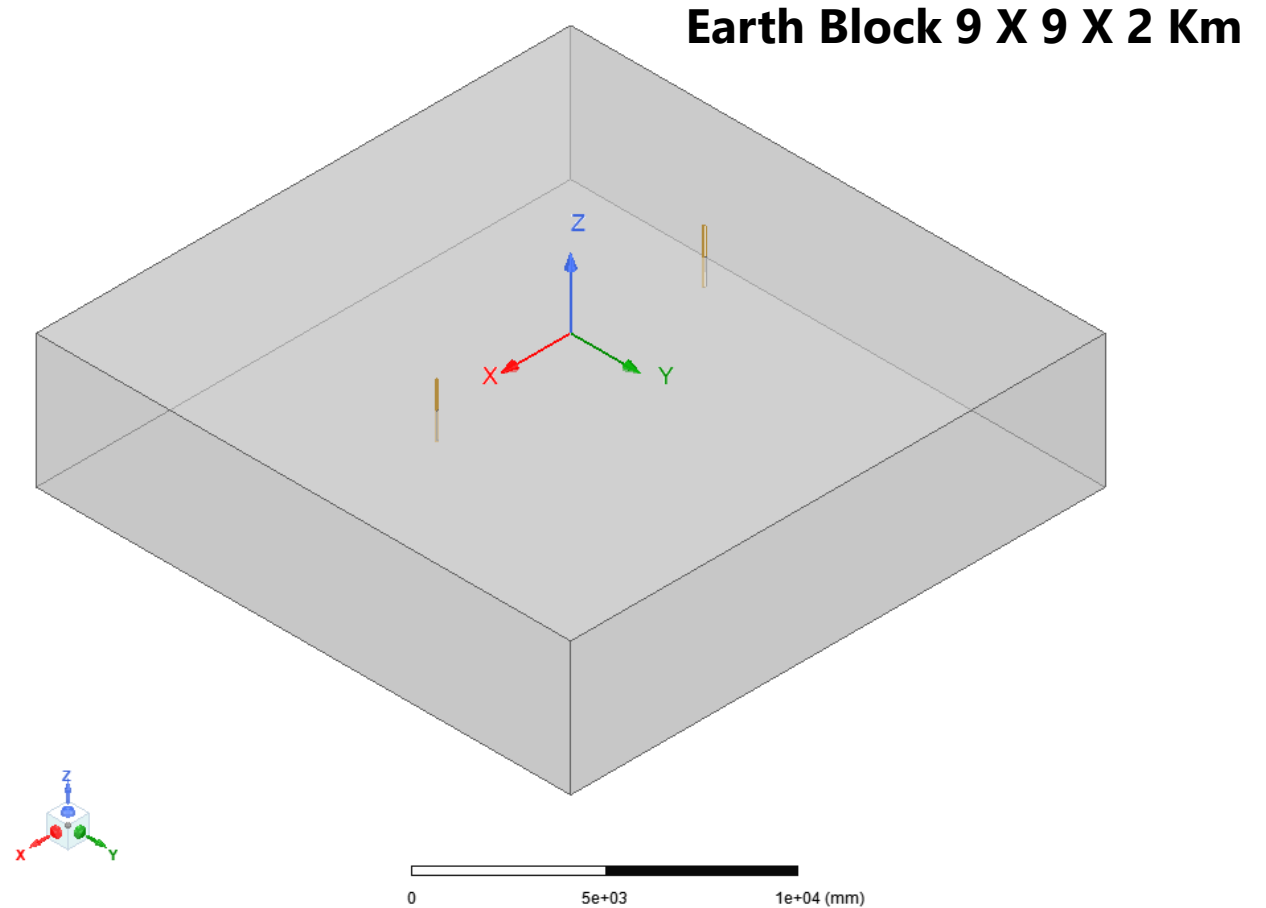
Simulations

Software Tools

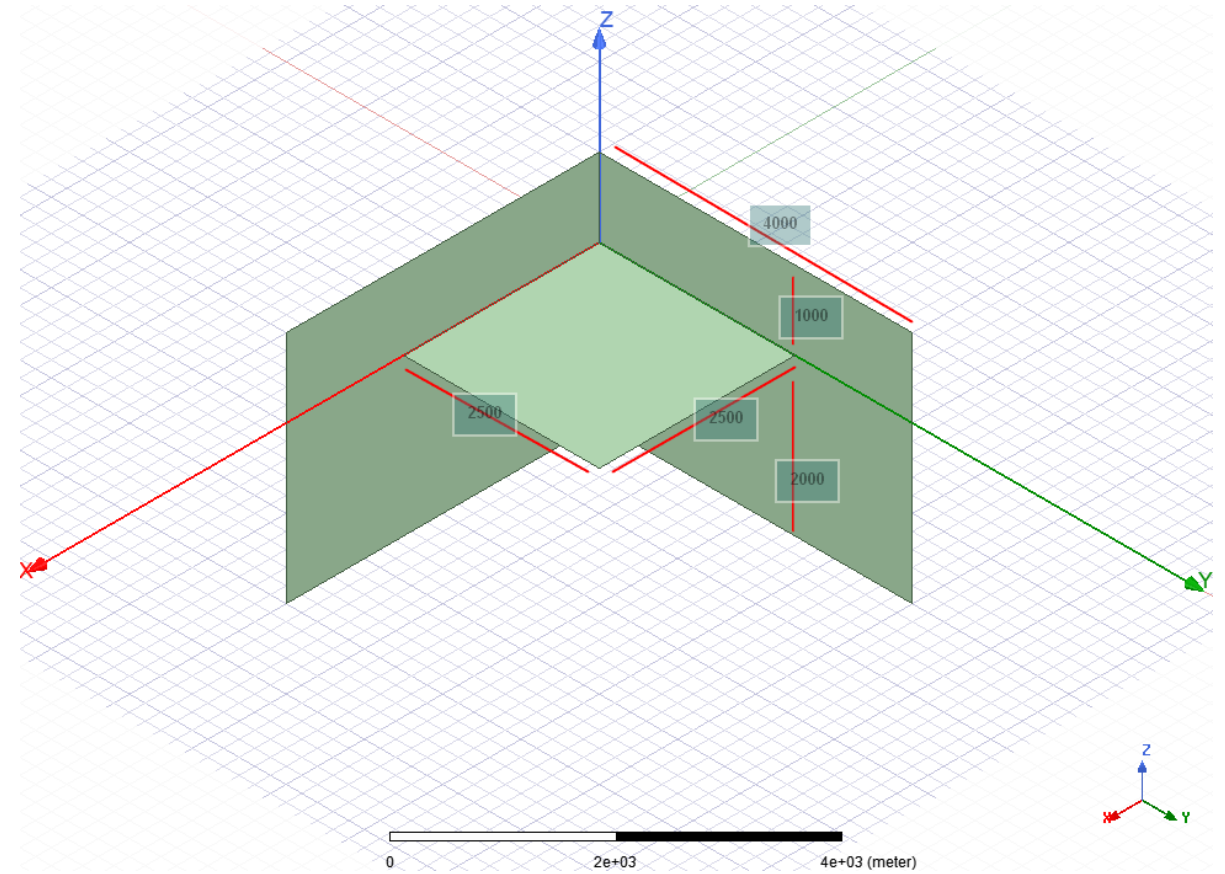
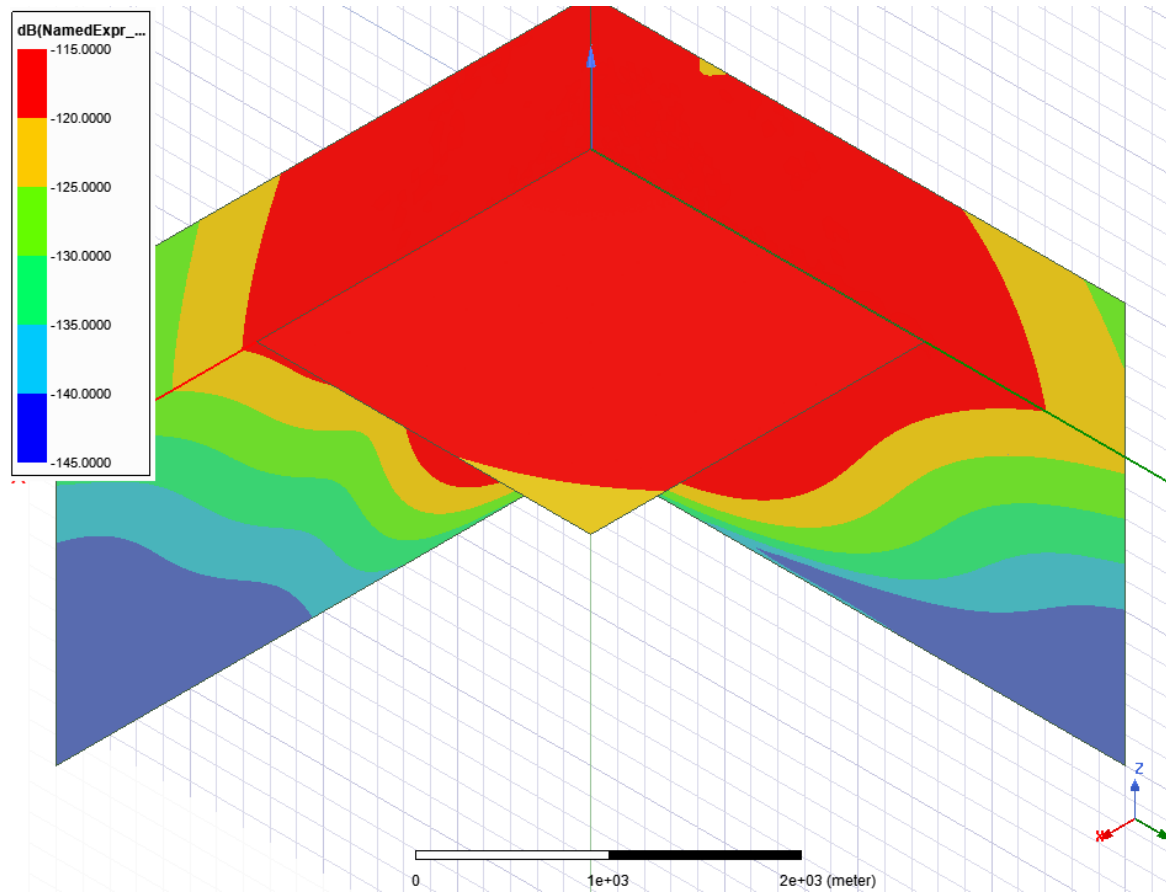
- ANSYS HFSS 3D Electromagnetic Simulation Software
- RES2Mod
- CST Studio (3D EM analysis & Design)

Simulation Values

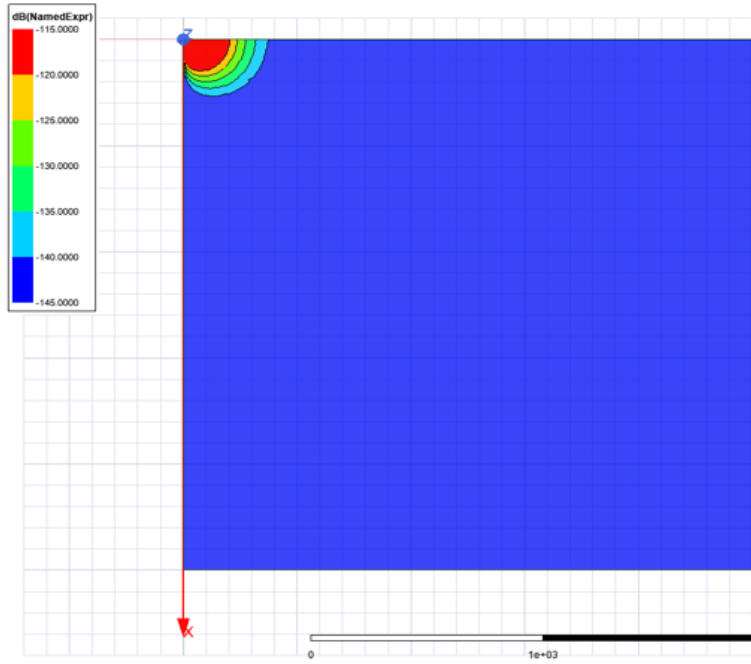
TX Electrodes: Metal $200 \times 6 \times 6 \text{ cm}^3$ 1 meter deep
Conductivity: $\sim 0.01 \text{ S/m}$ (wet soil)



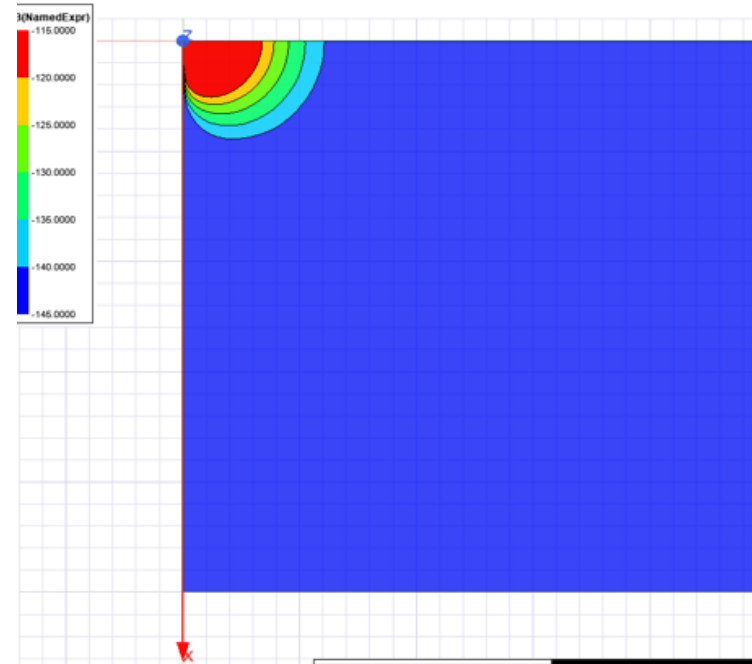
Earth mode



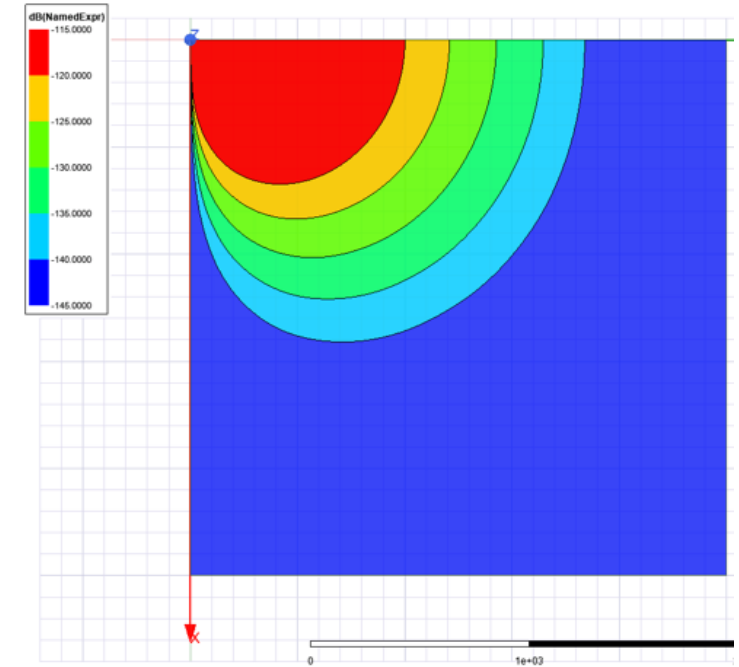
TX Transmission Simulations



Simulation 1: 73 kHz



Simulation 2: 9 kHz



Simulation 3: 0.1 kHz

TX Electrodes separation ~10m

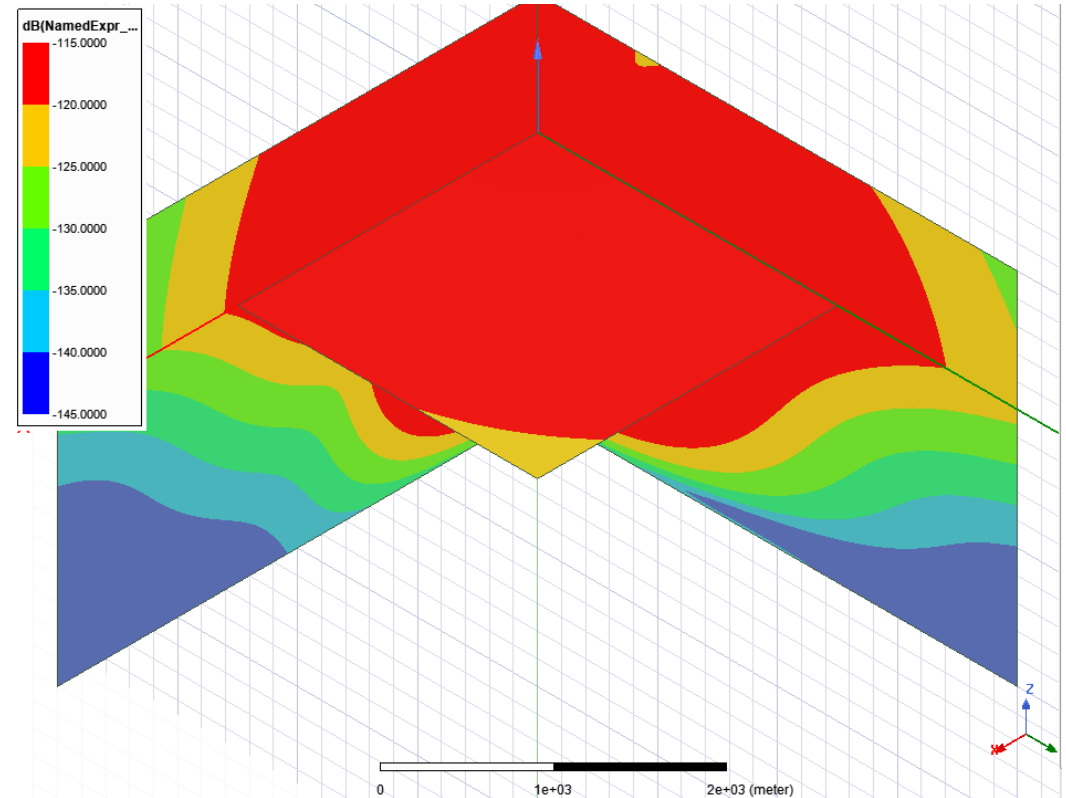
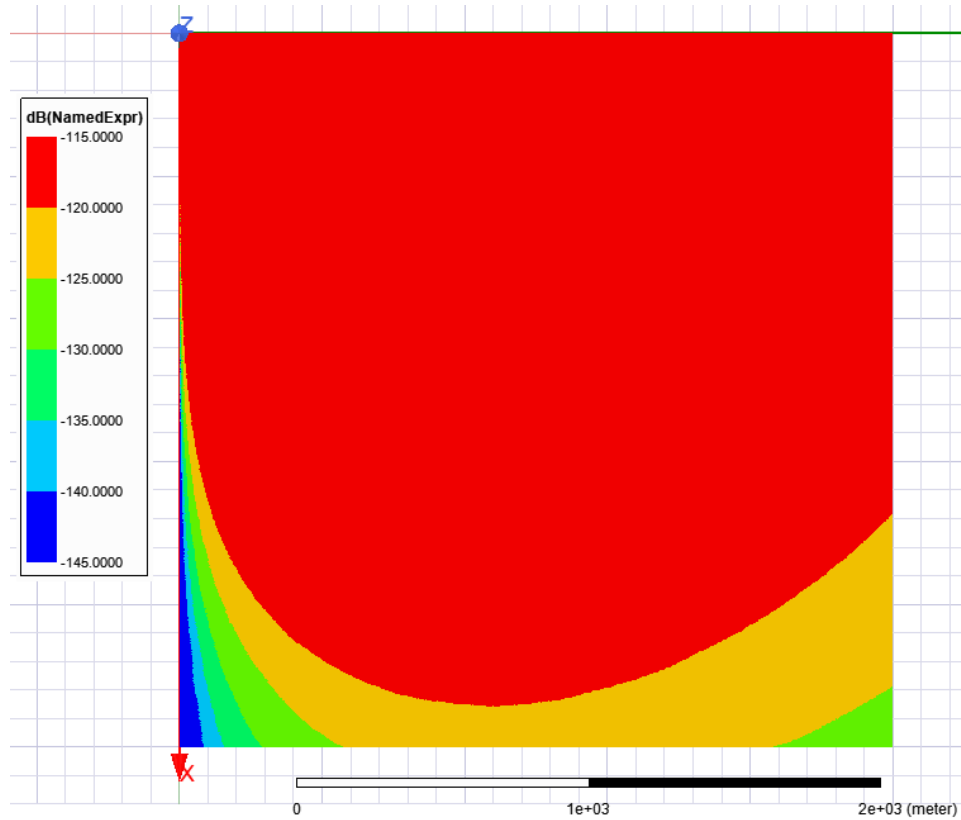
Injected power ~ 1A

Distance scale 100m/div.

Initial Random Frequency Results

Frequency (kilohertz)		Electrode Separation (Metres)	Injected Power (Watts)	Operational Range (Metres)
0.1		~ 10	~ 3600	~ 1900-2400
0.1		~ 10	~ 1800	~ 1800-2200
0.1		~ 20	~ 3600	~ 2600-3200
0.1		~ 20	~ 1800	~ 2600-3000
0.1		~ 50	~ 3600	~ 2800-3500
0.1		~ 50	~ 1800	~ 2800-3500
0.1		~ 100	~ 3600	~ 3000-4000
0.1		~ 100	~ 1800	~ 2800-4000
0.1		~ 1000	~ 3600	~ 3800-4800
0.1		~ 1000	~ 1800	~ 4000-5000

TX Transmission Simulations



Frequency: 0.1 kHz
Electrodes: 100 Meters
Power: 3600 Watts

Noise (Natural and Man Made)

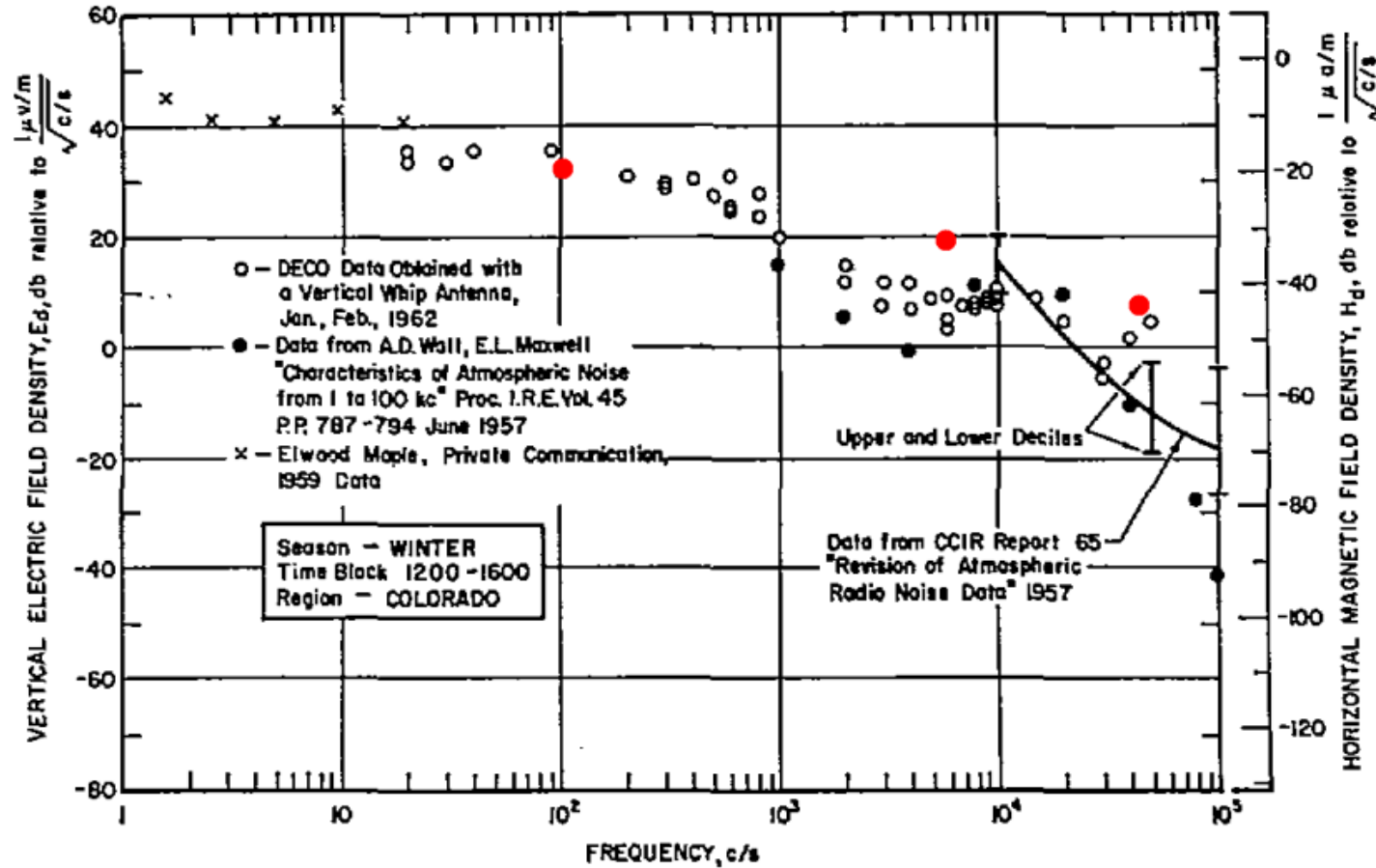


Fig. 1—Natural noise density.



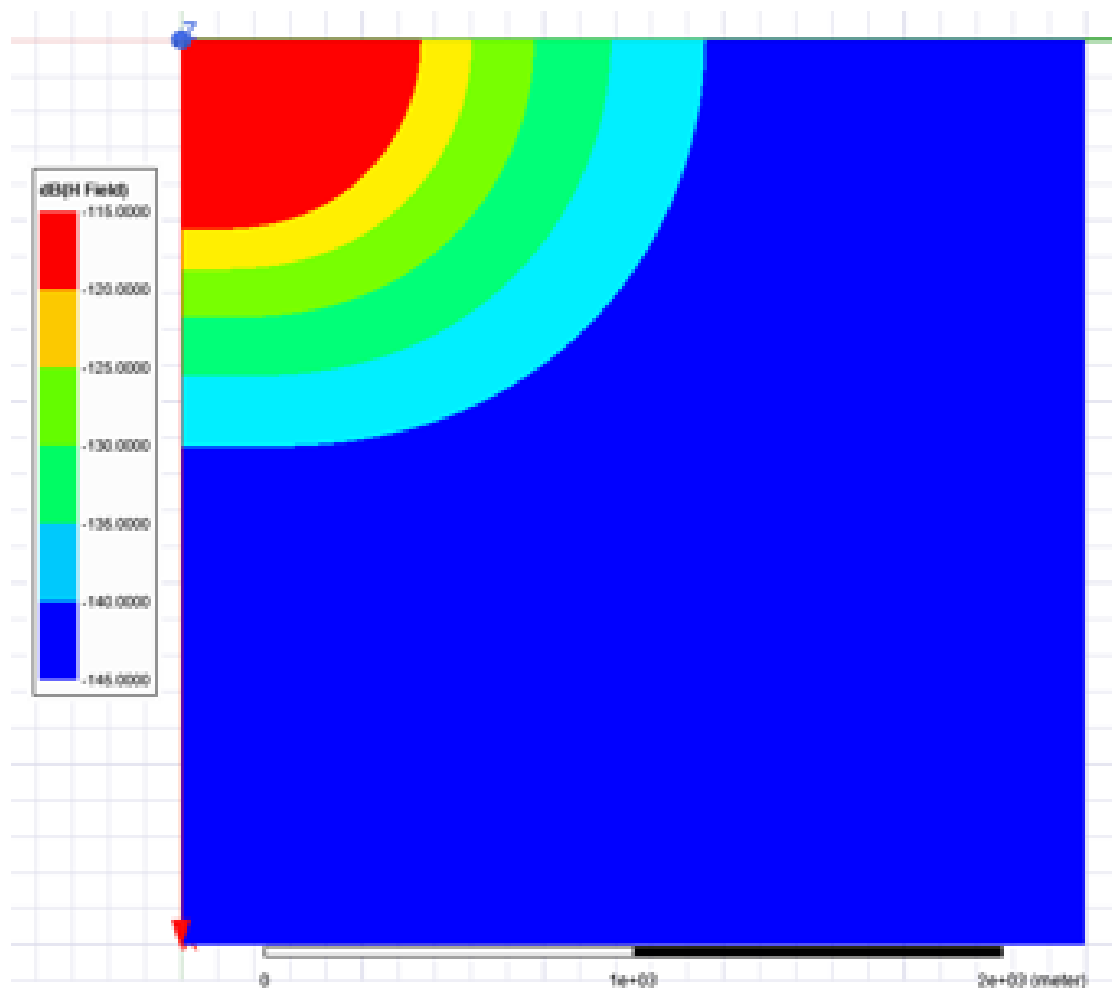
Ideal Range

Frequency: 1-4 kHz

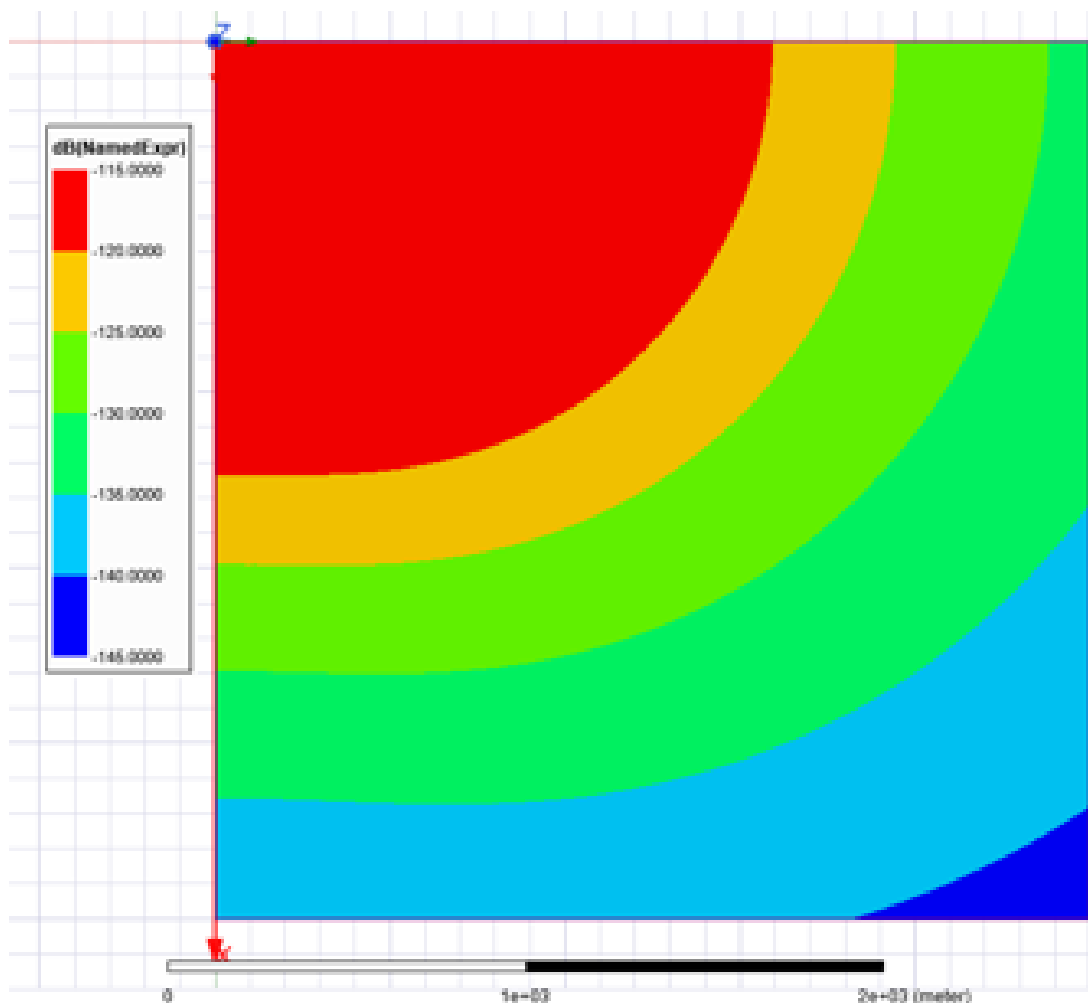
Electrodes: 10-100 Meters

Power: 3600 Watts

Low Power vs High Power

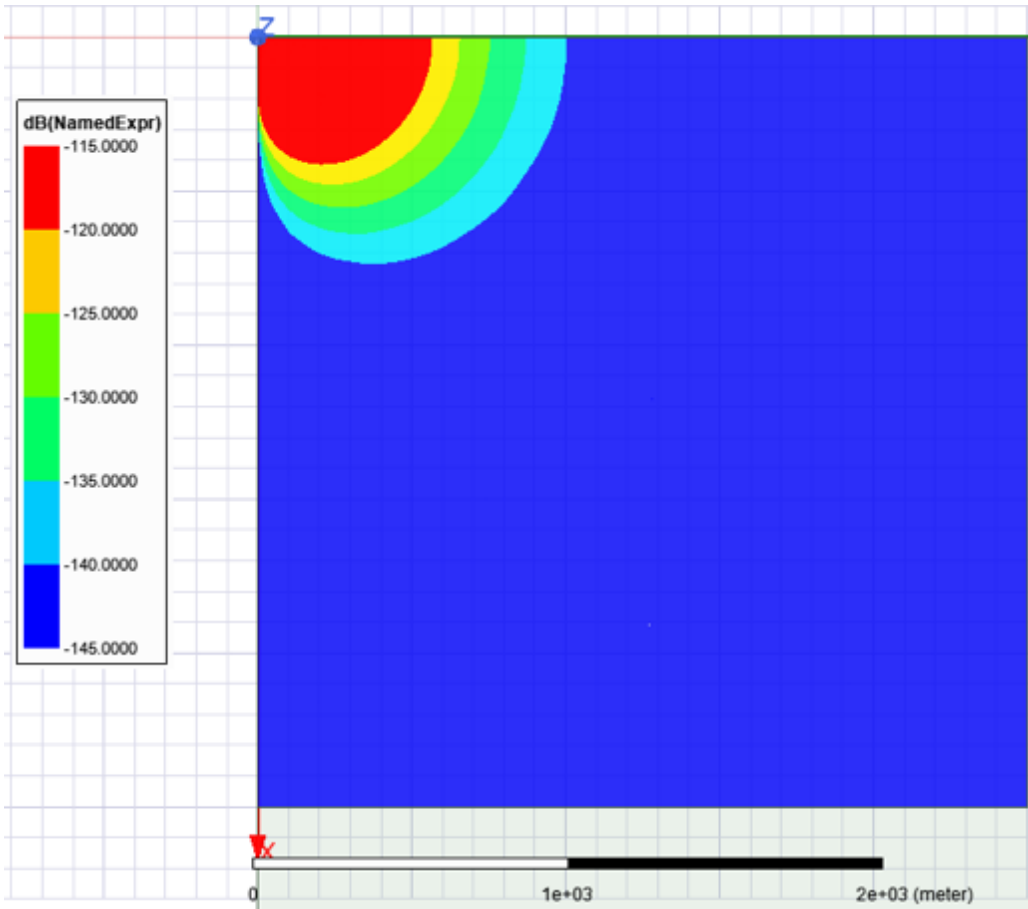


20W, 9kHz,
20-meter Electrodes
Distance **~1km**

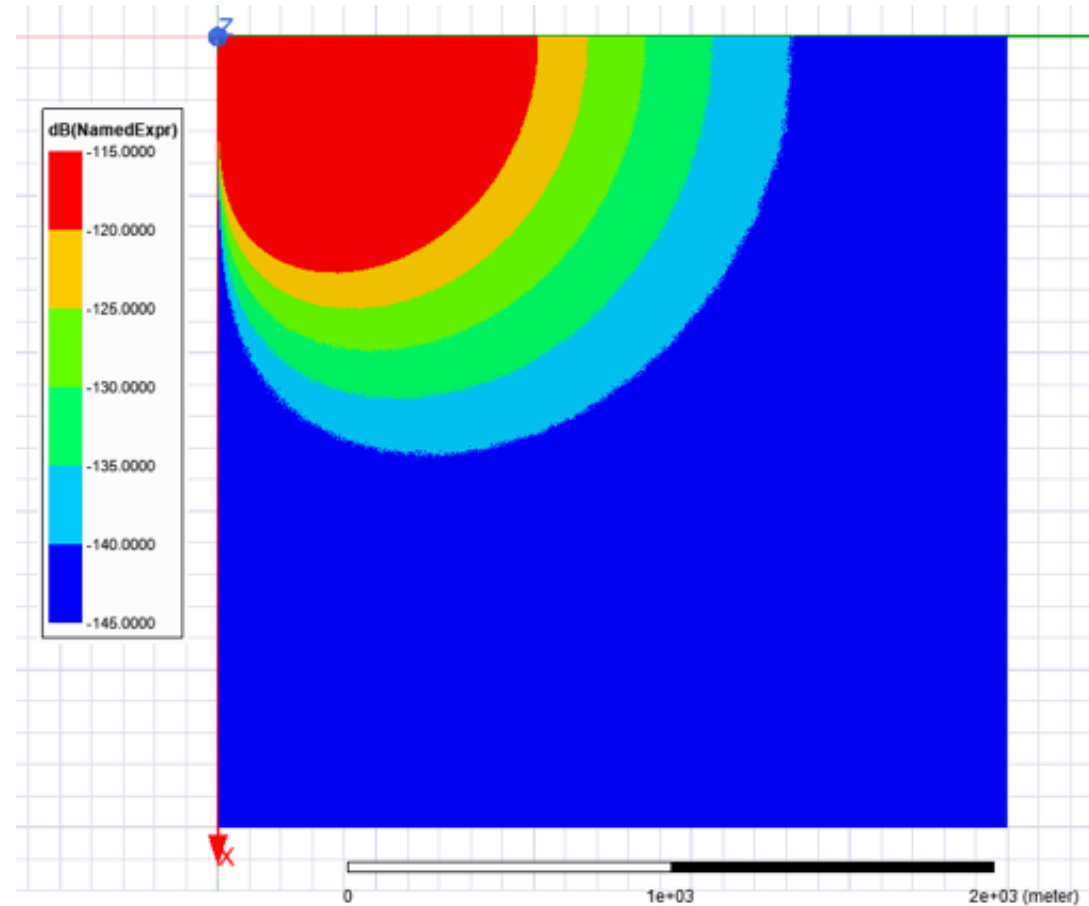


3600W, 9kHz,
20-meter Electrodes
Distance **~2.5km**

Tx Electrode Separation

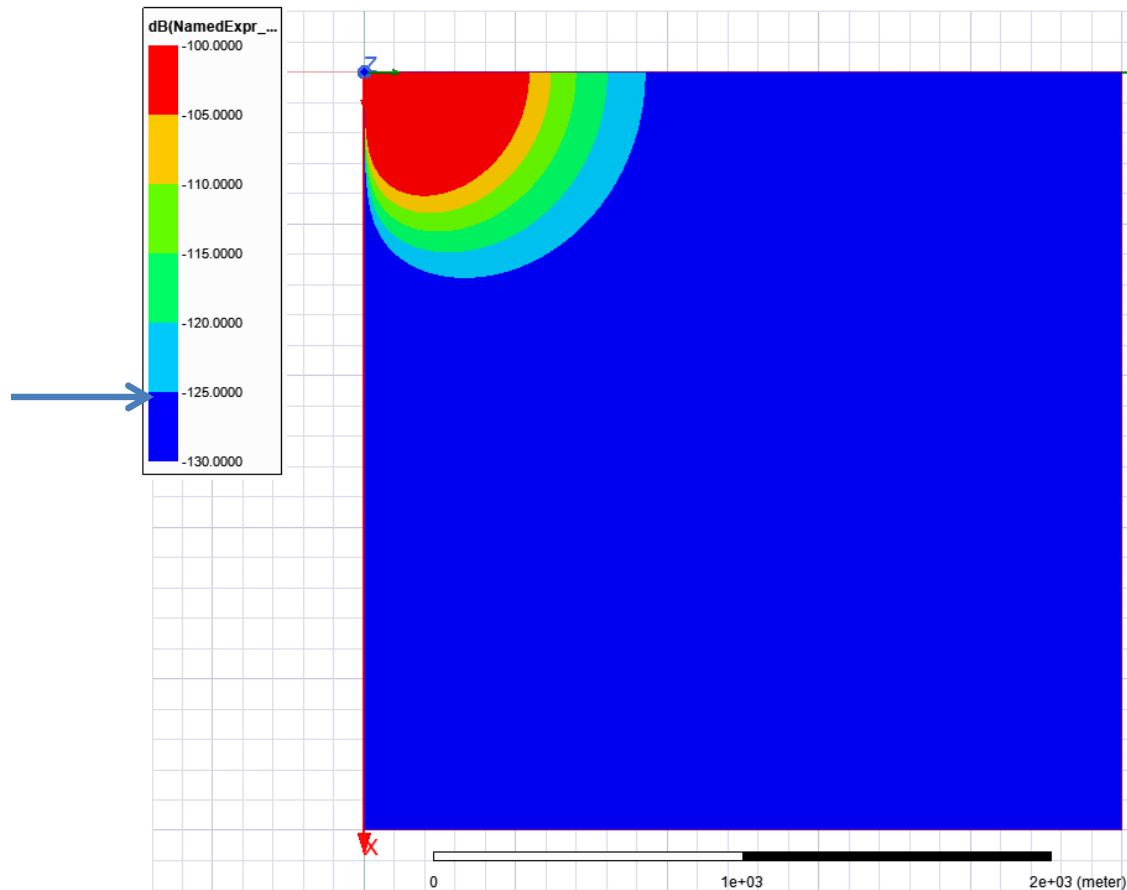


3600W, 9kHz,
10-meter Electrodes
Distance **~1km**



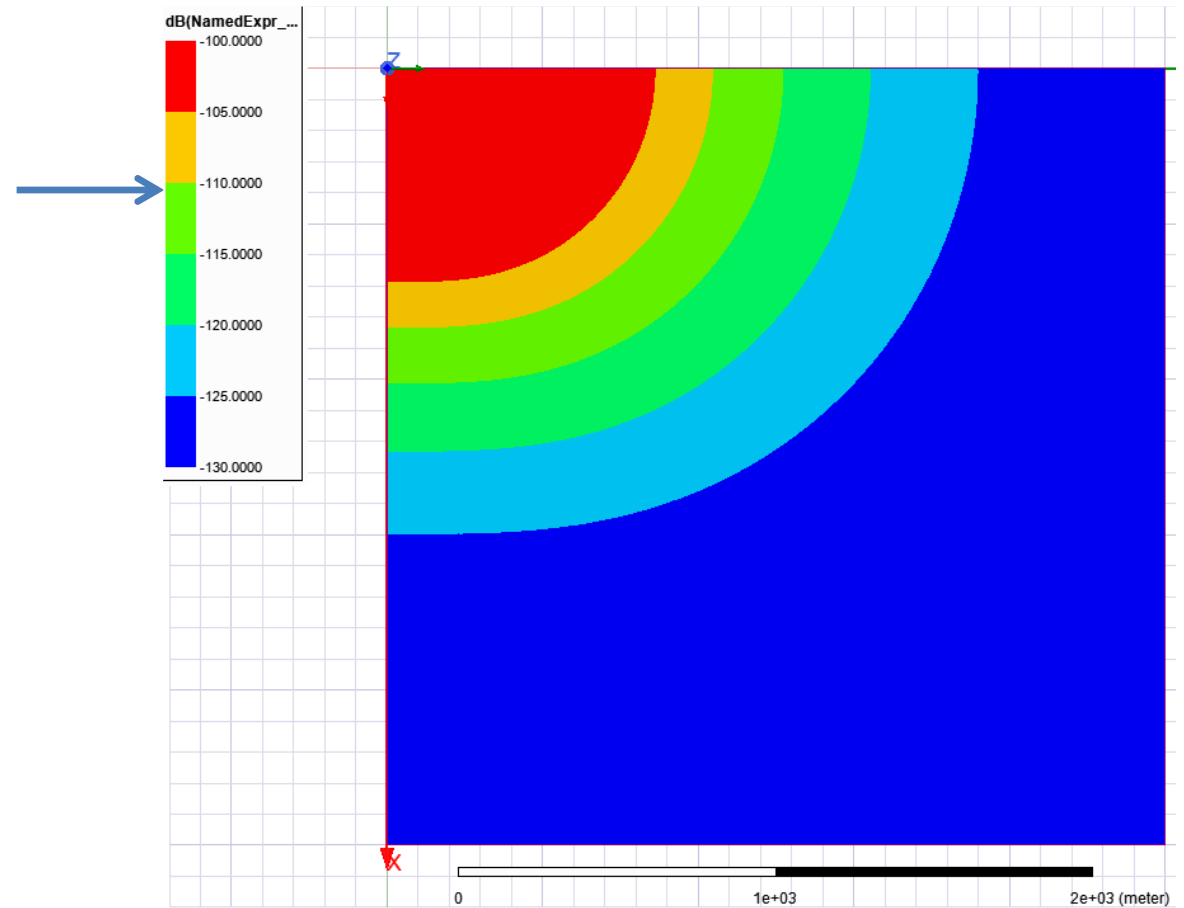
3600W, 9kHz
100-meter Electrodes
Distance **~1.6km**

Normal vs Tangential Simulation



Normal

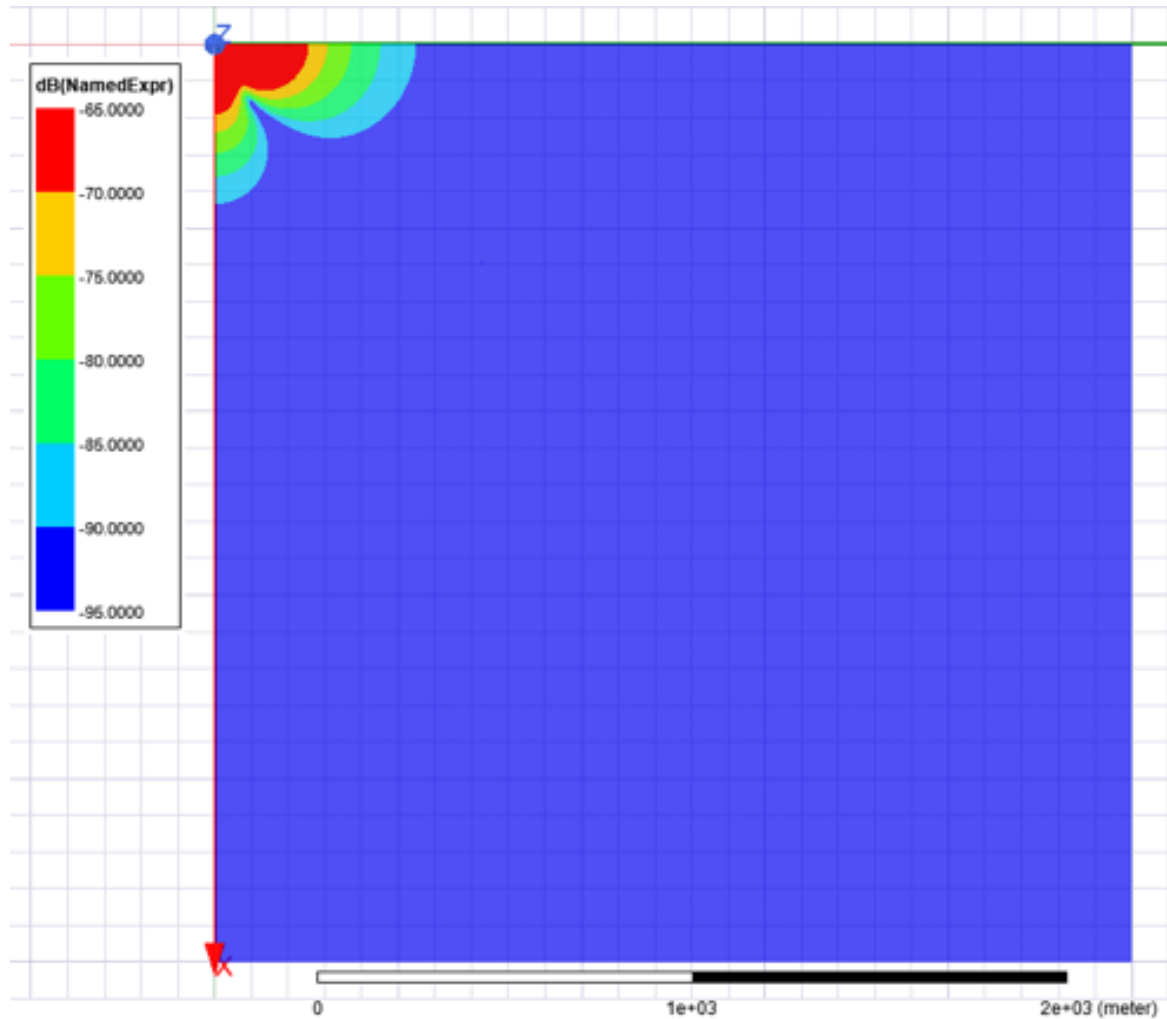
(laying down)



Tangential

(Vertical loop)

E- Field Tests



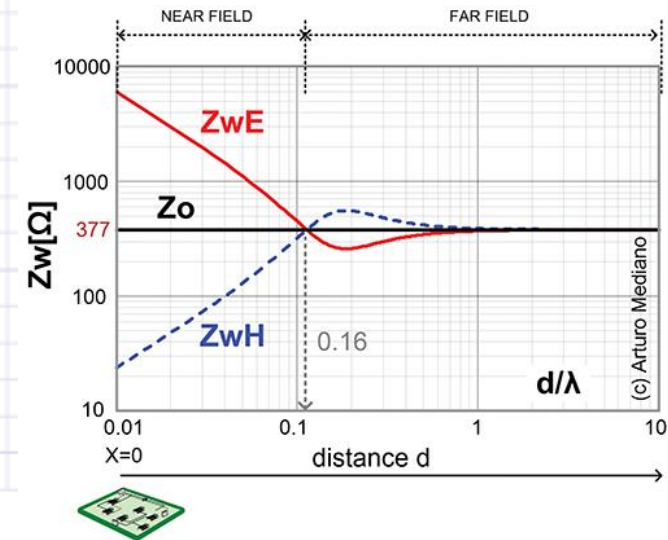
Frequency: 9 kHz

TX Electrodes separation $\sim 10\text{m}$

RX Electrodes separation $\sim 10\text{m}$ aligned X-direction

Injected power $\sim 3600\text{W}$. Accepted power $\sim 3560\text{W}$.

Distance scale 100m/div.



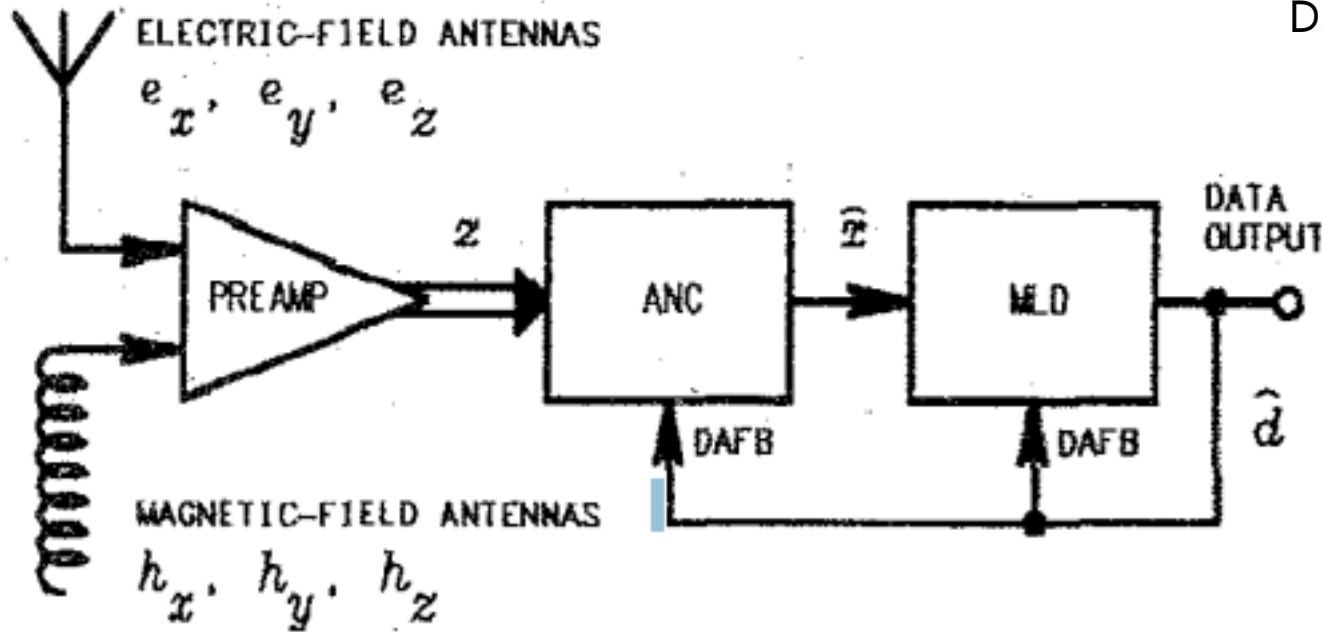
Impedance Matching

Earth Conductivity S/n	0.0001 (Dry)	0.001 (Sand)	0.01 (Soil) Reference Case	0.1 (Clay)
Matched Impedance	Normal: 2km Tangent: 1.5km Source impedance: ~6000 ohm	Normal: 1.9 km Tangent: 1.85 km Source impedance: ~600 ohm	Normal: 1.4 km Tangent: 1.85km Source impedance: ~100 ohm	Normal: 1.05 km Tangent: 1.9 km Source impedance: ~7 ohm
Non matched Impedance	Normal: 1.45 km Tangent: 0.7 km Source impedance: ~100 ohm	Normal: 1.75 km Tangent: 1.65 km Source impedance: ~100 ohm		

Modulation Techniques

– SNR (Signal to Noise Ratio)

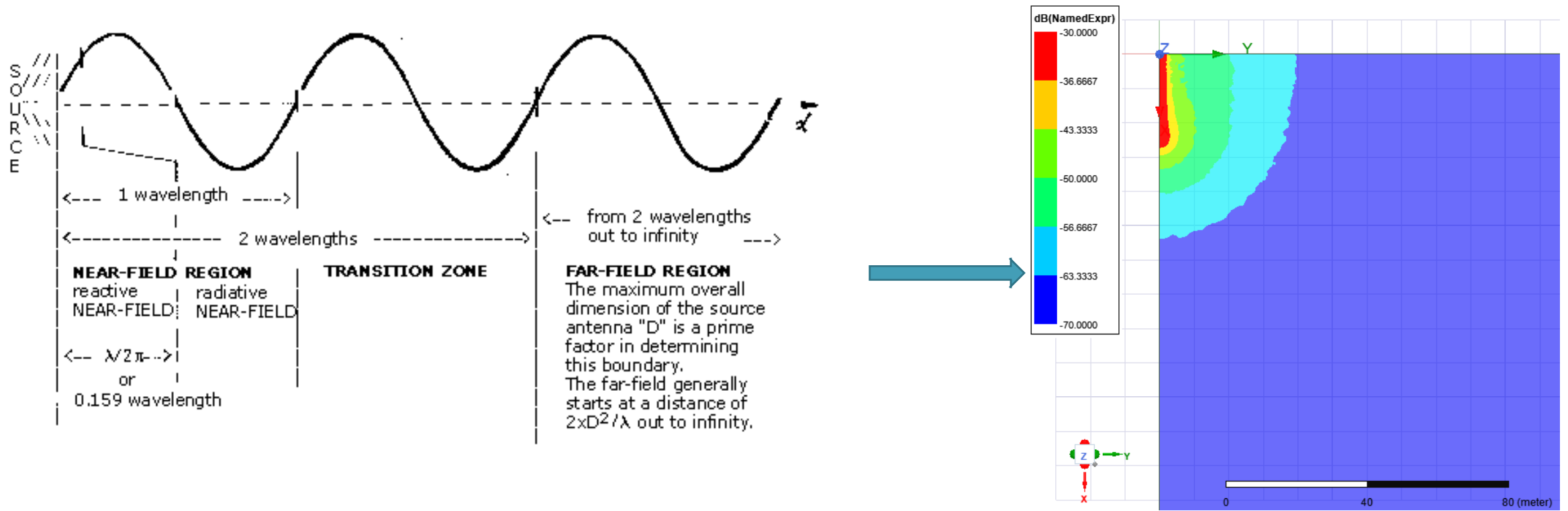
ANC: Adaptive Noise Cancellation
MLD: Maximum Likelihood detection
DAFB: Decision Aided Feedback



Modulation Choice improves SNR by 10-30 dB in TTE
(doubling operation distance)

* Paper includes Schematics for FSK, ASK and MSK modulation.

POE – Power through the Earth



The estimation of the USB DC power supply capability by means of Wireless Power Transfer (resonant NFC coupling) is explained below with simulations.

To provide power **P** at a perfectly matched load (USB powered device, for example) the required strength of magnetic field **H** is: **~50 meters**

$$H = \frac{4 \cdot \sqrt{\rho}}{f \cdot \pi^{3/2}} \cdot \frac{\sqrt{P}}{\sqrt{S_w} \cdot D^{3/2} \cdot \sqrt{N}}$$

Where:

D – diameter of Rx coil (assumed 1m)

N – number of turns of coil winding (assumed 100 turns)

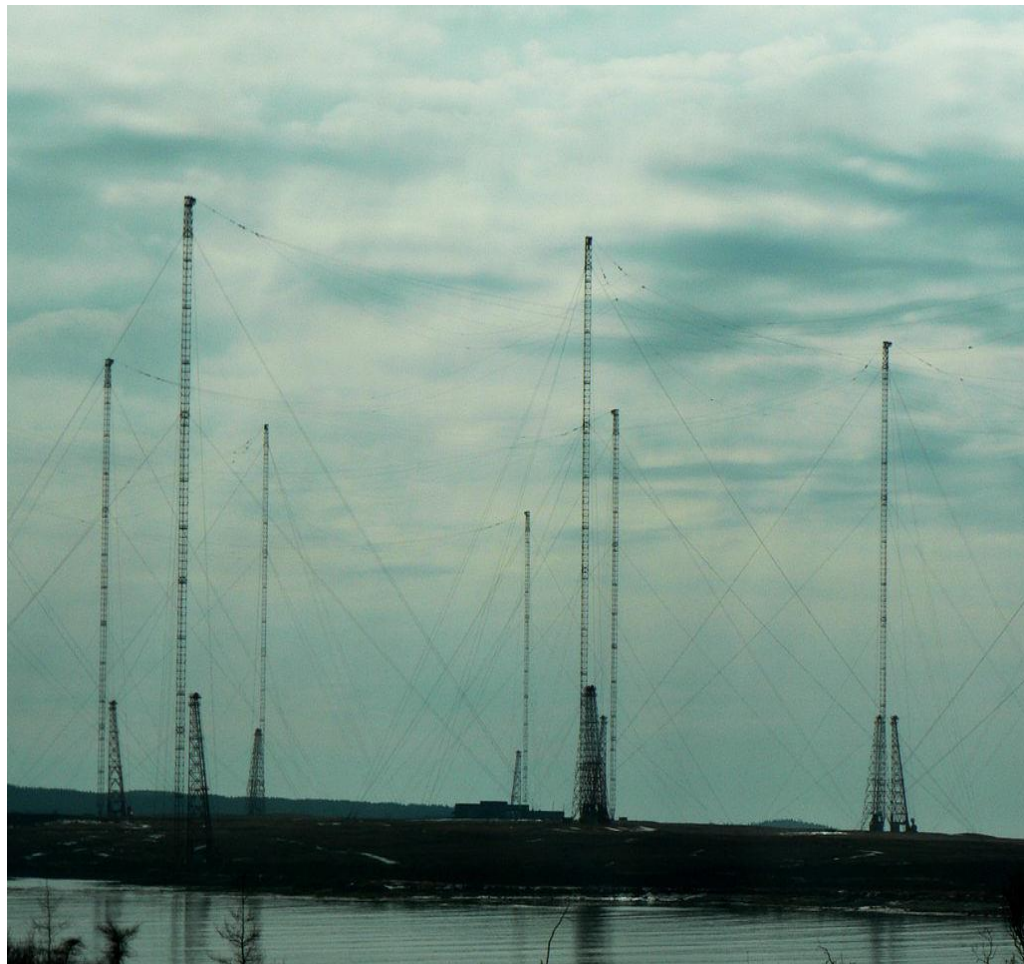
S_w – cross-section area of wire (0.312 mm²)

f – frequency of wireless power transfer (2 kHz)

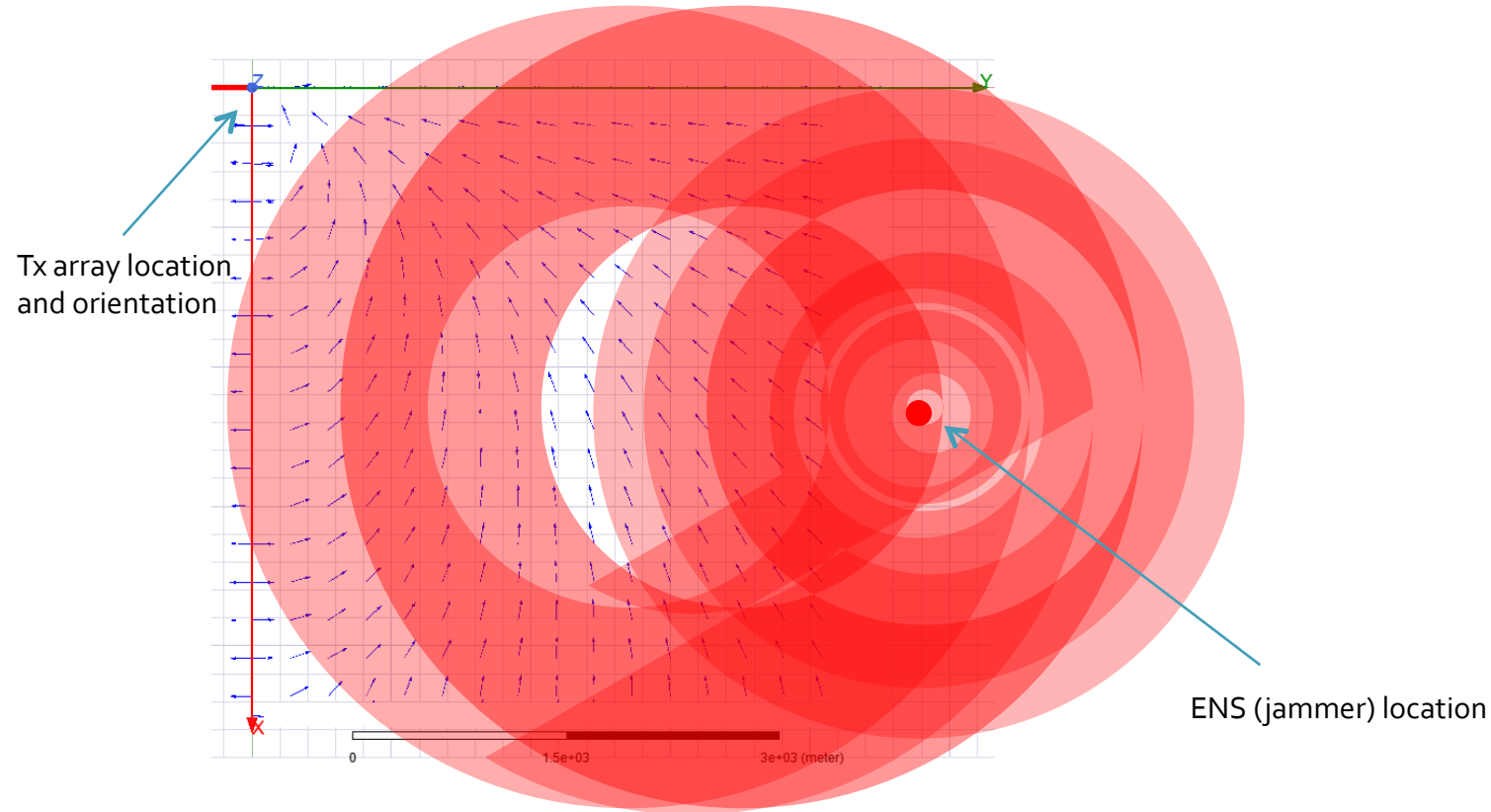
p - resistivity of wire conductor (1.72 x 10⁻⁸ Ωm)

The estimated required strength of magnetic field **H** is -62 dBA/m

Antenna / Near Field / Far Field



Tangential vs Normal

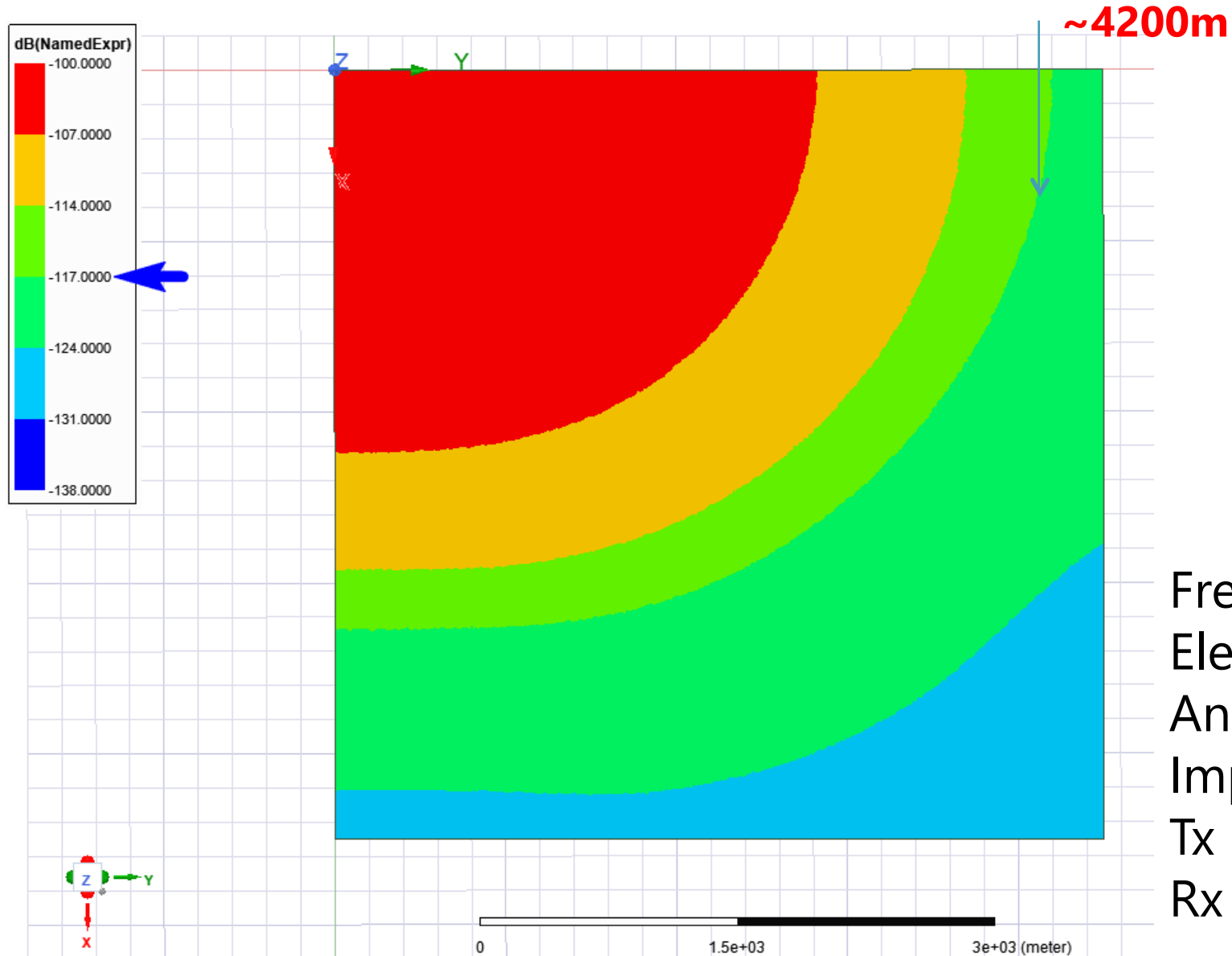


Normal: Loop Antenna Flat on ground is not affected

Tangential: Is Affected, Required Multi – Rx right angle Vertical Loops

** More details in paper

Playing with the Variables



Frequency 2 kHz
Electrodes: 100m separation
Antenna Tangential
Impedance: Perfectly Matched
Tx power: 20 kW
Rx sensitivity: 5kg Copper Wire

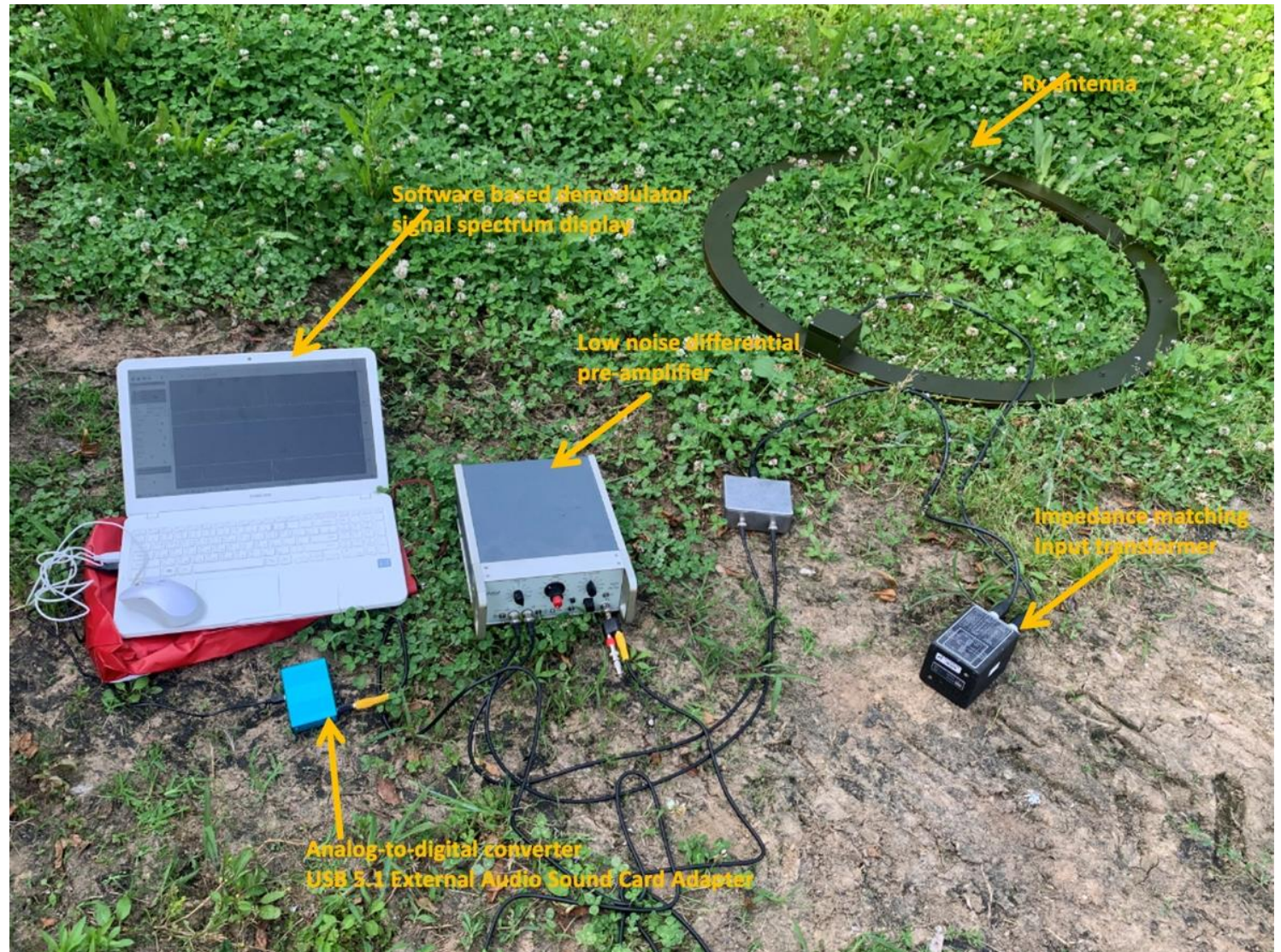
VLF/ULF – Earth Mode Hardware

**ECCM – Electronic Counter
Countermeasure**

Custom Tx/Rx Prototype



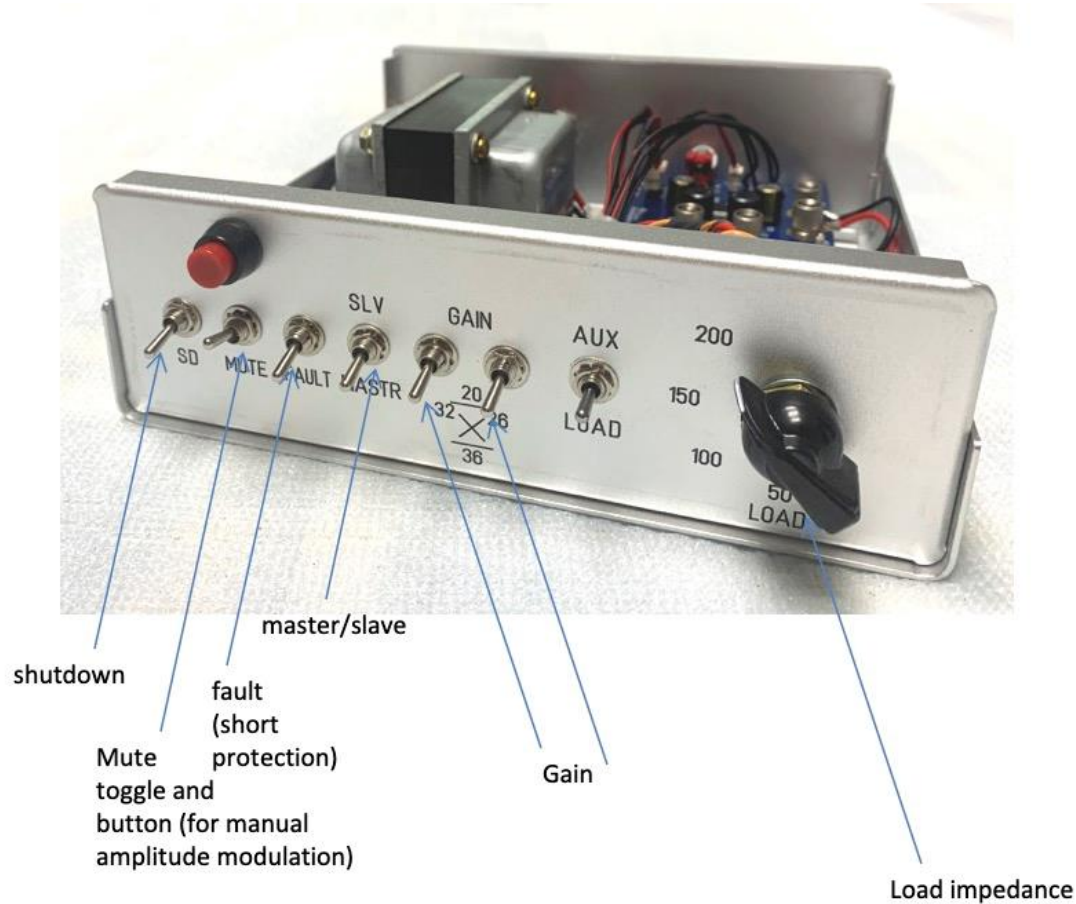
Tx



Rx

Custom Transmitter

Prototype



Commercial Version



Smaller Tx Containing

- **20W+3600W Power Amp**
- **Amp to Tx Antenna Matching**
- **Arduino/Pi (Data post processor/logger/encryption)**
- **LCD Display**

Custom Rx

Smaller Rx Containing

- Rx Antenna Coil
- Band Pass Filter (BPF) input TX matching impedance
- Low Noise Pre-Amp
- Arduino/Pi (Data post processor/logger/encryption)
- LCD Display

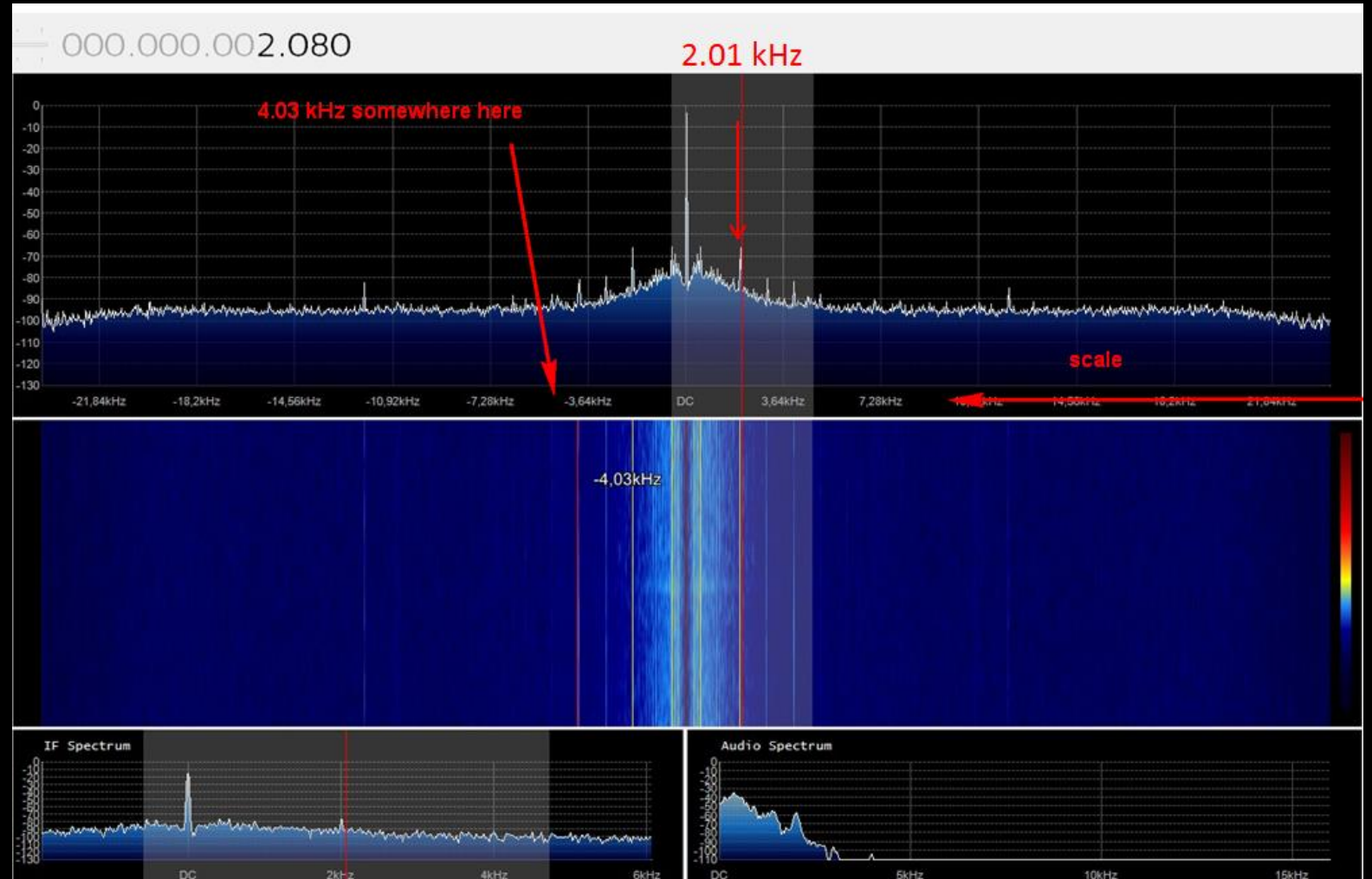
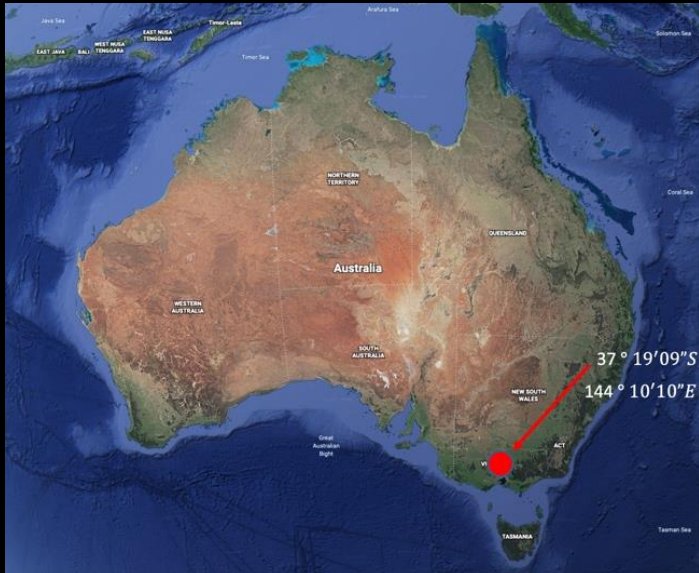


Electrodes and Antenna



TX Electrodes 1metre Steel Pipe

~2 kHz Test Results



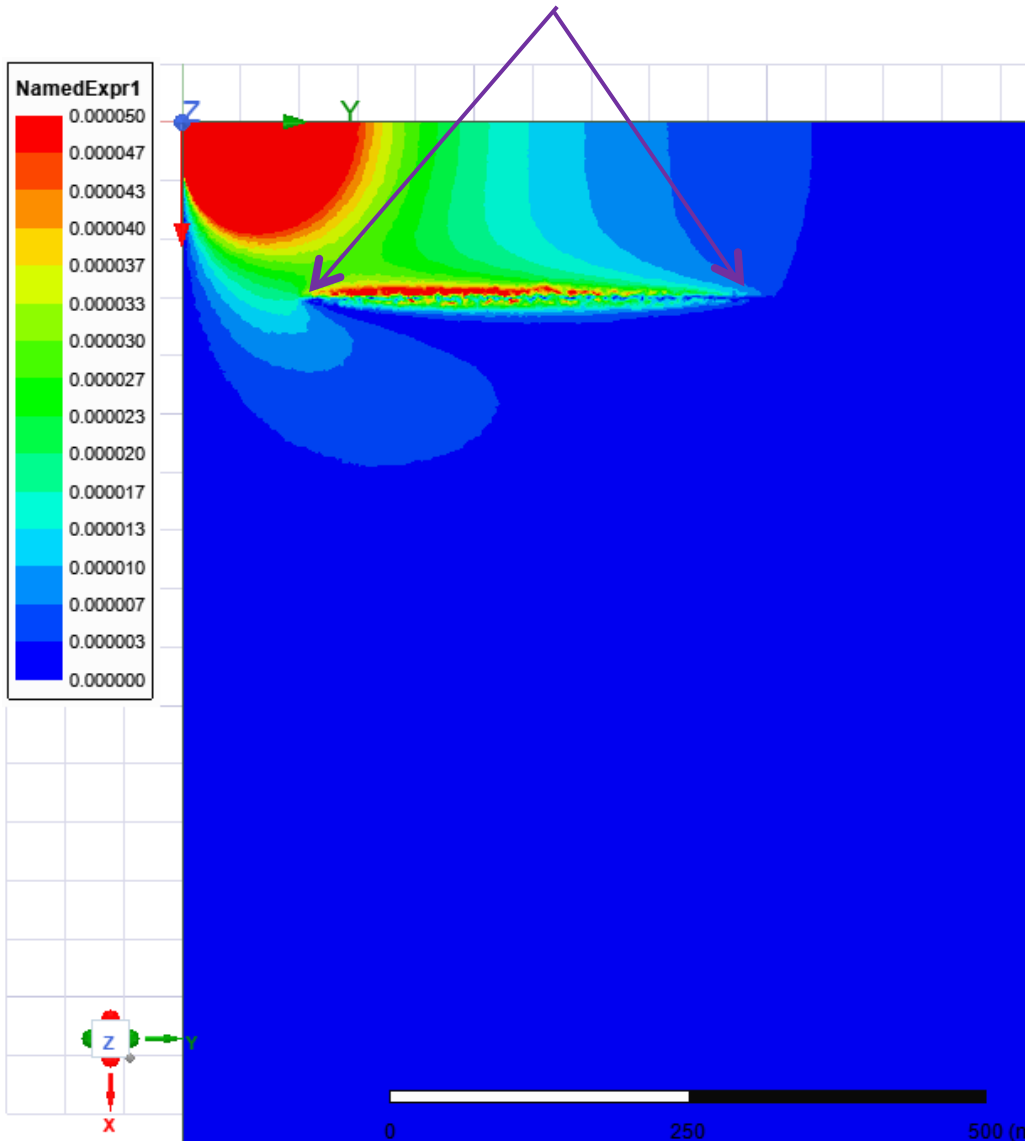
TX: Electrodes 50 meters
Power: **20 W**
Impedance: 100 Ohm
Modulation: **ASK**

RX: 1 meter diameter coil
Impedance: 100 Ohm
Preamp: AC, LF roll-off
– 1kHz, HF roll-off – 1kHz,
Gain – 200 (linear).

Distance: 923m

Utilities Assisted Mode

The ends of the fence wire here



VLF – Utility Assisted Mode

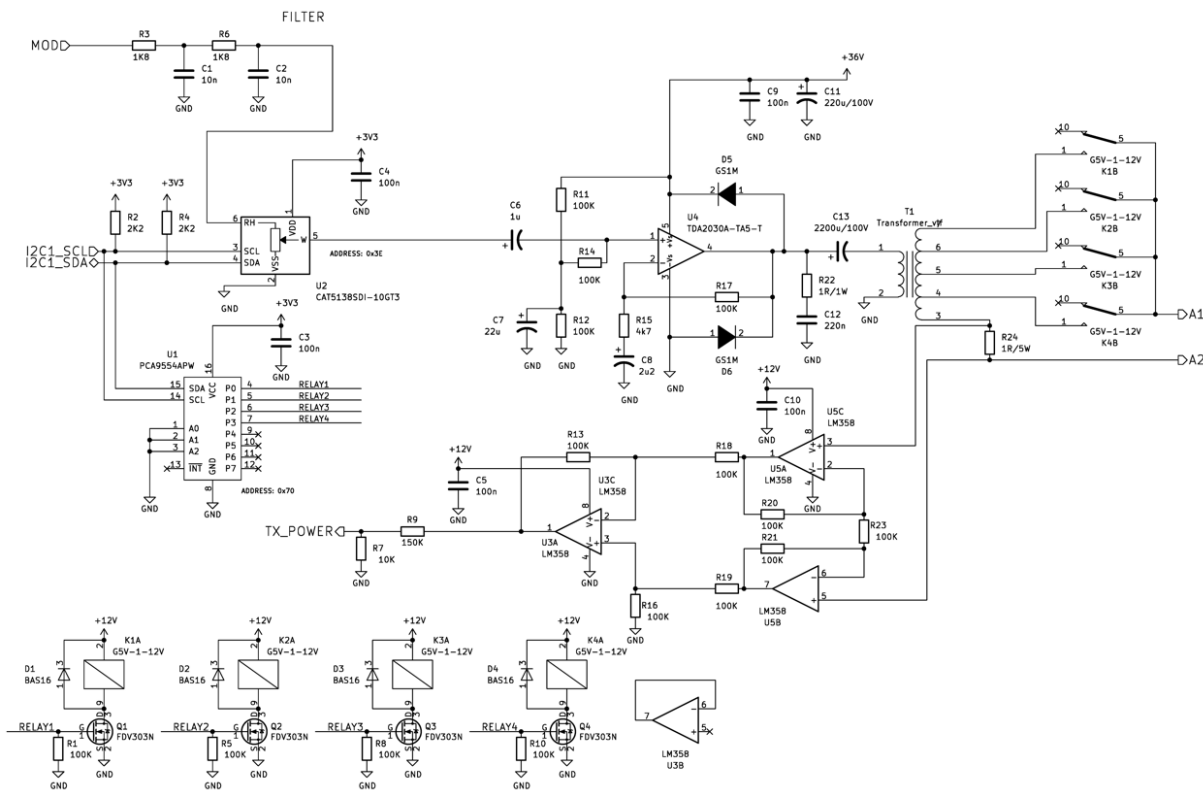
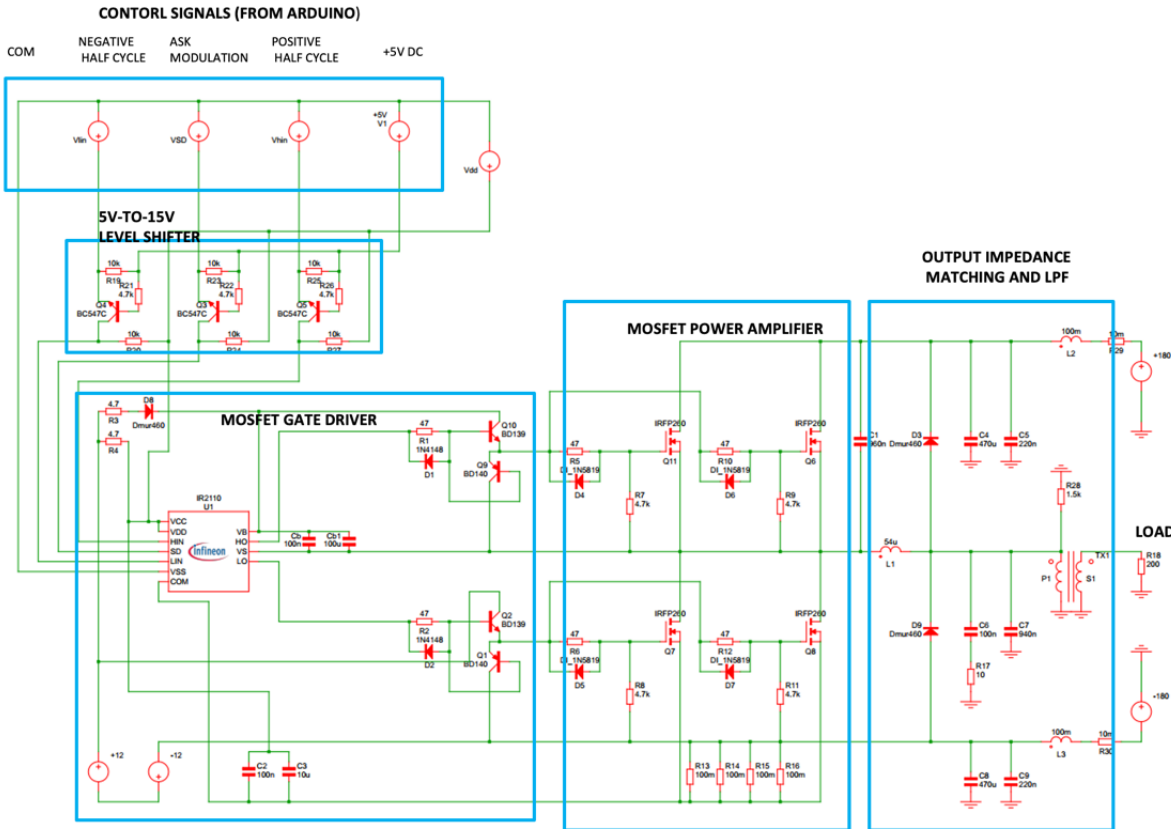
Hobbyists have recorded 2-10km of increased signal detection with underground steel pipes, wires, etc.

***Will impact Urban distances positively

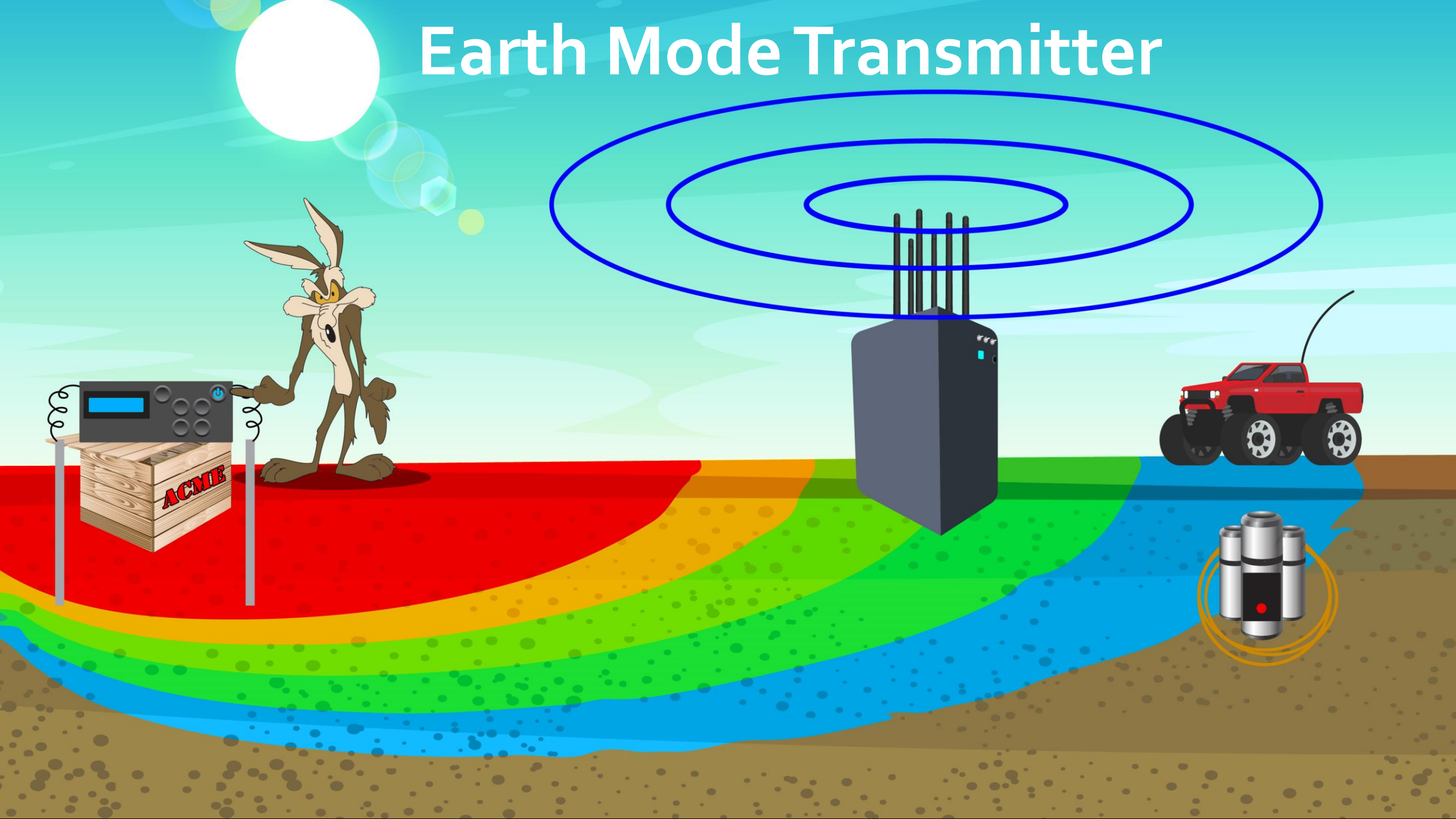
Home Brew

Stage	Materials	Costs	Comments
Base band processor (Arduino)	Arduino Board	\$20	Message encoded for Arduino
Tx pre-amplifier (modulator)	Wire, jumpers, switches, modulated with TX ~20W Amp. Can be removed with TX fully synthesized digital source with Arduino above.	\$300 / \$0	An off the shelf Tx ~20W PA design kit
Tx ~20W Power Amplifier	Enclosure, DC 24V 7A*h rechargeable battery pack	\$100	Cutting, drilling, PCB Assembly, and soldering.
Tx amp-to Tx antenna matching	Copper wire, transformer core and coil former	\$50	Transformer winding and soldering
Tx Antenna (electrodes)	~55mm ID Steel Pipe 2m ~50mm OD Steel pipe 2m 200x800x3 mm^3 steel plate Bolts and welding clamps	\$115	Cutting, griding, bending, welding, and soldering.
Rx antenna (coil)	3 pcs of 1000X1000X10mm ^3 Foam PVC sheet. Copper wire, enclosure, connector, PVC bond	\$170	Cutting, bonding, winding, soldering.
Rx BPF and impedance matching or amp below	Low noise input impedance matching transformer or pre amp below (either or)	\$2500 new or \$200 custom	Rare stock and overpriced new, Replaced with a custom narrow band transformer for \$200
Rx low noise pre-amplifier	Differential input low noise preamplifier	\$1000 new or \$100 custom	Rare stock and overpriced can be replaced with a custom Dual Amp building block chip. (LT1567 Dual ABB) or Preamp LNA10
BPF amplifier and signal conditioner	PCB, components	\$50	PCB Assembly
Data post processor and data logger	Arduino Board	\$20	Message encoded for Arduino

Home Brew



Earth Mode Transmitter





Use Case



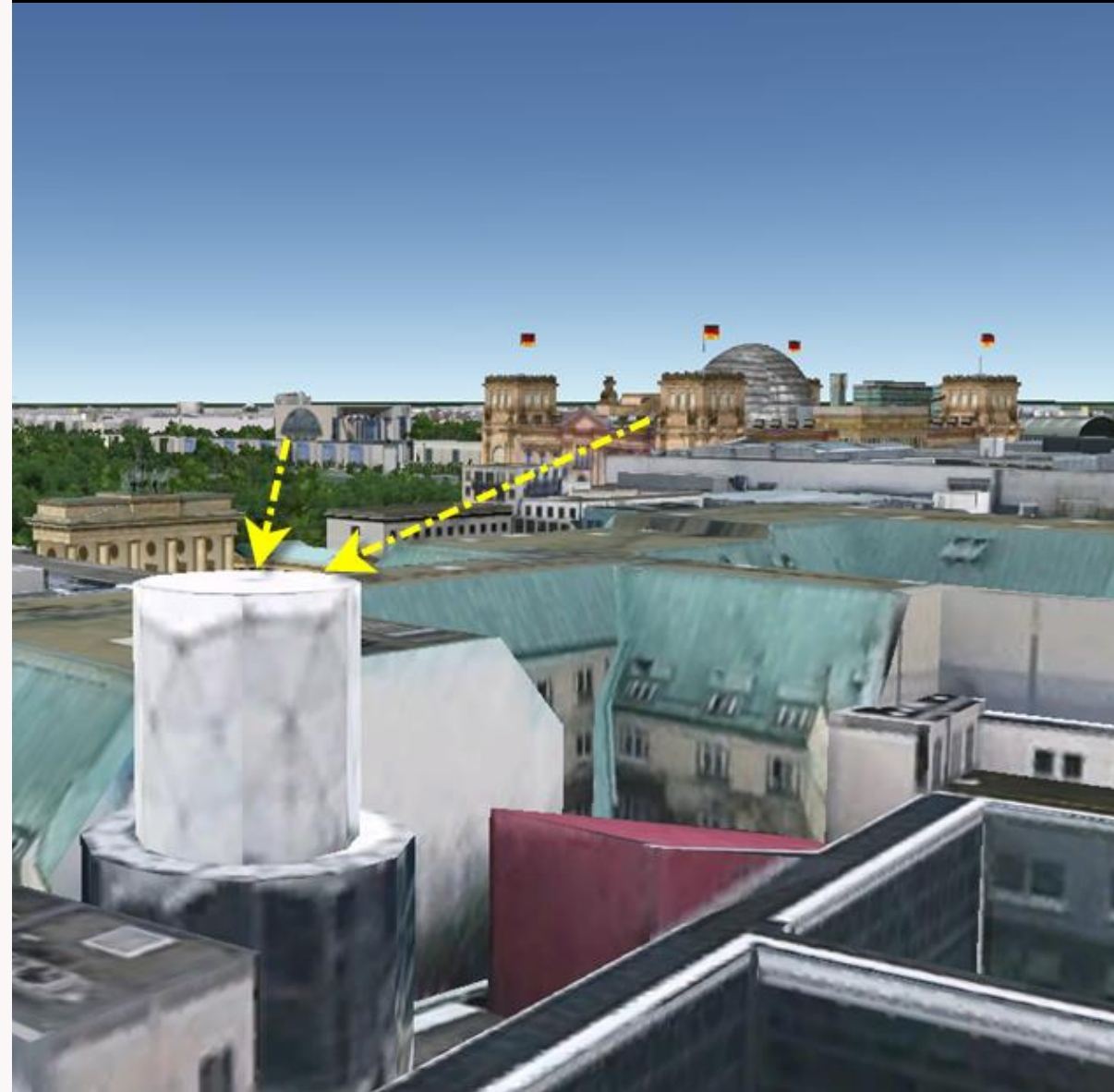
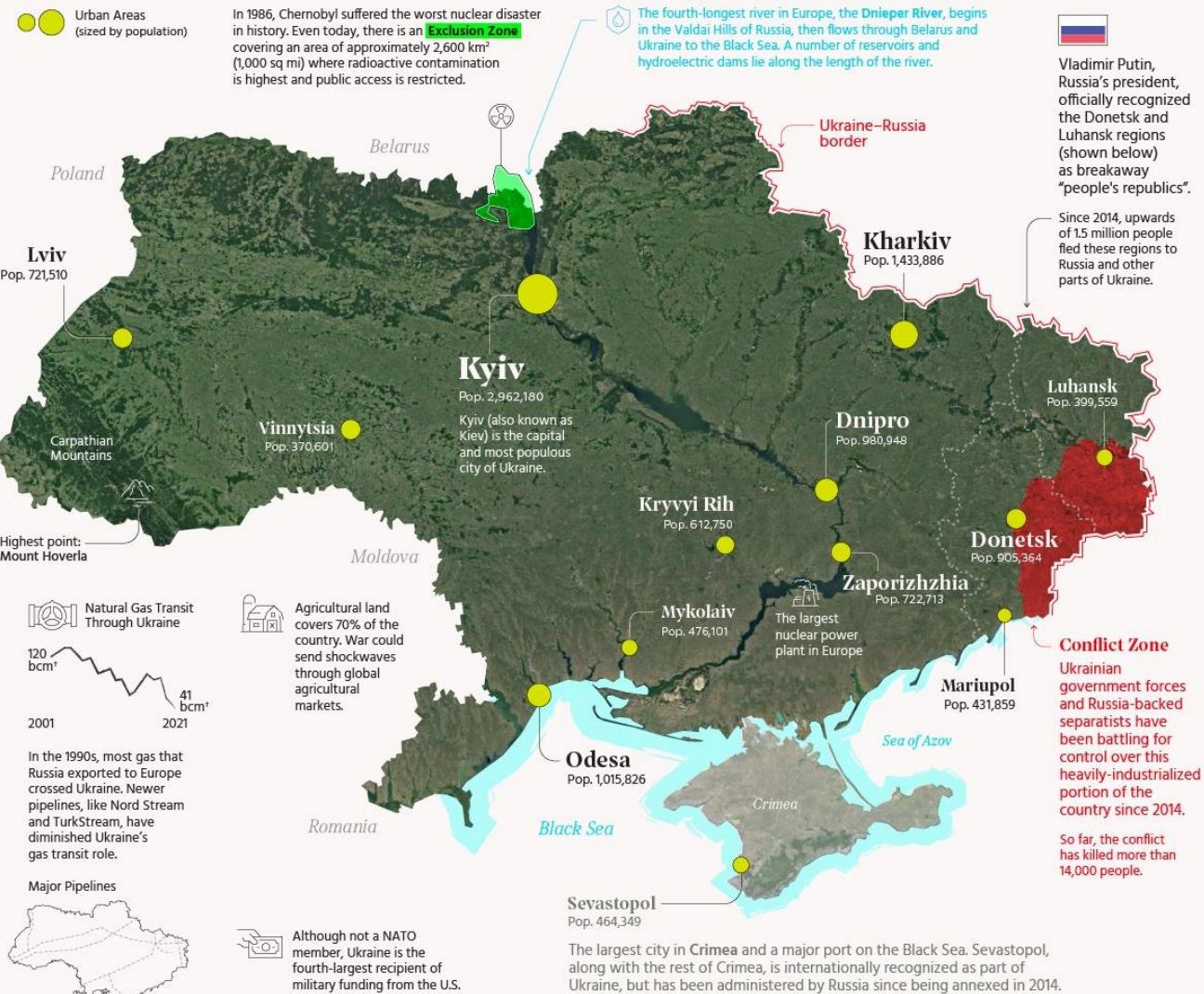
Use Case

MAP EXPLAINER Ukraine

GDP (nominal)	\$155.6B	54 of 206	Land Area**	603,628 km ² (233,062 sq mi)
Population	43,467,779*	35 of 206	Pop. Density	~75/km ² (196/sq mi)

*Excluding the autonomous Republic of Crimea and the city of Sevastopol

**About 7% of Ukraine's land area is Russian occupied

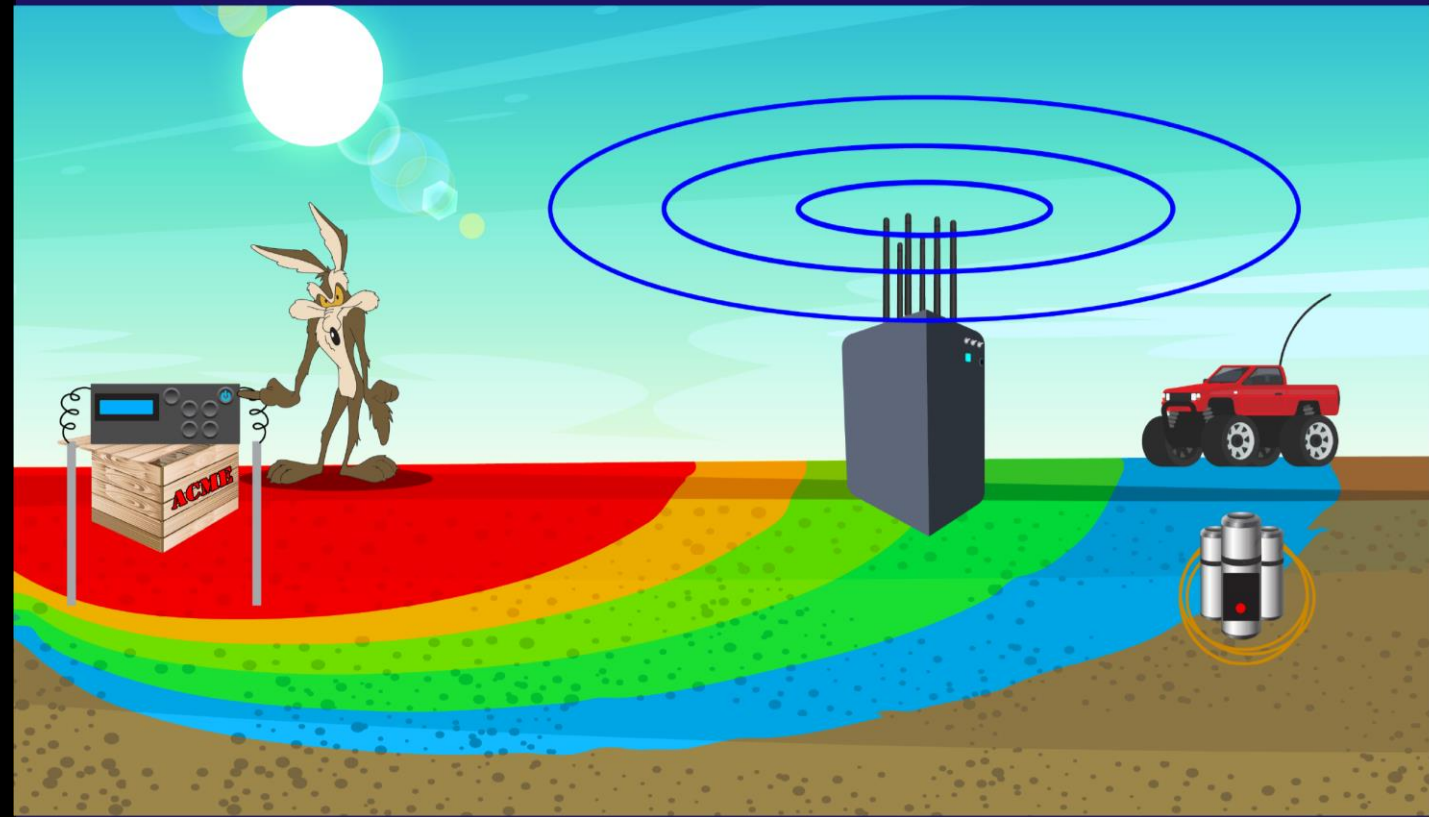


Summary

- **Send messages up to 11km (7 Miles)**
- **Operates at 1- 4 kHz**
- **Conductive Current Magnetic field ULF**
- **Runs in Near Field Magnetic Field (H Field) not Far Field.**
- **TX operate from 20Watts to 3600W power scales to 20 kW**
- **RX runs on 20Watts**
- **Impedance Matching Options**
- **Encrypted messages using Libraries**
- **Modulation options**
- **Jammer proof**
- **Small factor RX Arduino/Pi or can use Laptop PC for messages**

The End

CONDUCTIVE CURRENT EARTH MODE COMMUNICATIONS



Radio Frequency Jammer
Electronic Counter Countermeasure